

An
Industrial Training Report
of
Laytime Calculator

Submitted to

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By

Shalin Nilesh Bhavsar (202003103510056)

Under the Guidance
of

Mr. Rushabh Shah

AM/NS
INDIA

Department of Web Development
AM/NS House,
27 km Surat-Hazira Road, Hazira,
Surat - 394270
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With immense pleasure, I present this report on the “**Laytime Calculator**” project.

I am grateful to my guides **Mr. Rushabh Shah, Mr. Ansh Agarwal, and Mr. Zaid Pathan** at **Arcelor Mittal Nippon Steel India Ltd. (AM/NS India)** for giving me the necessary support and encouragement that was necessary for the completion of this project.

Shalin Nilesh Bhavsar (202003103510056)

ABSTRACT

The Laytime Calculator Application represents a significant improvement in streamlining vessel operations for Arcelor Mittal Nippon Steel India Ltd. (AM/NS India Ltd.). Manual laytime calculations previously caused inefficiencies and errors, leading to the development of this innovative solution. This application automates the laytime calculation process, offering an easy-to-use interface and real-time calculations. It seamlessly integrates with a relational database, ensuring data safety and enabling Demurrage report generation. Key features include dynamic stage addition, customizable percentages, and connectivity with other databases for enhanced functionality.

The application's impact is significant, enhancing operational efficiency, accuracy, and transparency in laytime calculations. It standardizes the process, reducing delays, errors, and disagreements while improving compliance with industry standards and legal requirements. Additionally, the application supports business intelligence efforts by using historical data for trend analysis and decision-making.

The implementation phase resulted in a robust application with a user-friendly interface, smooth data entry, real-time calculations, and support for business intelligence. Future improvements could include integrating with Account Payable processes for post-NFA payment processing, refining the user interface, and updating regulatory compliance.

In summary, the Laytime Calculator Application transforms vessel operations management, improving efficiency, accuracy, and compliance while facilitating informed decision-making through business intelligence.

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CHAPTER 1 INTRODUCTION

1.1 Overview

The Laytime Calculator Application has been developed to address the challenges faced by Arcelor Mittal Nippon Steel India Ltd. (AM/NS India Ltd.) in the laytime calculation process for vessel operations. AM/NS India Ltd. is a leading steel manufacturing company with a significant presence in the Indian market. The company operates multiple ports for importing raw materials and exporting finished products, making efficient vessel operations crucial for its operations.

Previously, laytime calculations at AM/NS India Ltd. were performed manually, leading to inefficiencies and errors in the process. The manual process was time-consuming and prone to human errors, resulting in delays and inaccuracies in calculating laytime for vessel operations. Moreover, the lack of a standardized system for laytime calculations made it challenging to track and manage laytime effectively.

Recognizing the need for a more efficient and accurate solution, AM/NS India Ltd. decided to develop the Laytime Calculator Application. The application was designed to automate and streamline the laytime calculation process, providing users with a user-friendly interface and real-time calculations. By implementing the Laytime Calculator Application, AM/NS India Ltd. aims to improve the efficiency, accuracy, and transparency of laytime calculations for vessel operations, ultimately enhancing its overall operational efficiency and competitiveness in the market.

1.2 Problem Definition

The manual process of calculating laytime for vessel operations at Arcelor Mittal Nippon Steel India Ltd. (AM/NS India Ltd.) posed significant challenges, including time-consuming paperwork, errors in manual data entry and calculations, lack of standardization leading to inconsistencies, limited transparency in tracking and monitoring laytime, compliance difficulties, and operational inefficiencies. These challenges resulted in delays, inaccuracies, disputes, and increased operational costs, highlighting the need for a more efficient and automated solution to streamline the laytime calculation process and improve overall operational efficiency.

1.3 Scope/Application

The scope of the Laytime Calculator Application includes:

- Allowing users to enter essential information, including vessel details, cargo specifics, NOR tendered date, commenced loading date, and completed loading date.
- Supporting multiple functionalities, including dynamic stage addition, date and time selection using a calendar, real-time laytime calculation, database integration, and data export for generating Demurrage reports.
- Integrating with a relational database management system (RDBMS) to store laytime calculation data securely.
- Allowing users to export laytime calculation data to an Excel file for recordkeeping and further analysis.
- Providing users with an intuitive graphical user interface (GUI) with interactive date and time selectors using a calendar to minimize data entry errors.

Applications

The Laytime Calculator Application developed for Arcelor Mittal Nippon Steel India Ltd. (AM/NS India Ltd.) has a wide range of applications within the company's vessel operations and logistics management. The application is designed to streamline and automate the laytime calculation process, making it more efficient, accurate, and transparent. Some of the key applications of the Laytime Calculator Application include:

1. Improved Efficiency:

By automating the laytime calculation process, the application improves the efficiency of vessel operations at AM/NS India Ltd. Personnel can quickly enter data, perform calculations, and generate reports, reducing the time and effort required for laytime calculations.

2. Accuracy and Transparency:

The application ensures the accuracy of laytime calculations by eliminating manual errors. It also provides transparency in the calculation process, allowing users to track and monitor laytime calculations in real time.

3. Standardization:

The application introduces a standardized system for laytime calculations, ensuring consistency and accuracy across different vessel operations. This standardization helps avoid misunderstandings and disputes with shipping companies.

4. Compliance:

The application helps AM/NS India Ltd. comply with regulations and contractual agreements related to laytime calculations. It ensures that laytime calculations are done under industry standards and legal requirements.

5. Data Analysis and Reporting:

The application allows users to export laytime calculation data for further analysis and reporting. Users can generate Demurrage reports and conduct trend analysis to improve decision-making and operational efficiency.

6. Business Intelligence:

The application serves as a repository for vessel-related data, enabling various departments, including bulk raw material, shipping, and chartering, to conduct business intelligence. Historical laytime records can be used for trend analysis, performance evaluation, and decision-making.

Overall, the Laytime Calculator Application enhances the overall productivity and competitiveness of vessel operations at AM/NS India Ltd. by streamlining and automating the laytime calculation process. It improves efficiency, accuracy, and transparency, while also facilitating compliance and business intelligence.

CHAPTER 2 TRAINING ACTIVITIES

During my training activities, I gained valuable experience and expertise in various technologies and skills. Here's a summary of my training activities and the technologies I learned:

- **Backend Development:**
 - Utilized Django framework for backend development.
 - Integrated PostgreSQL database for data storage and management.
 - Implemented CRUD (Create, Read, Update, Delete) functionalities.
 - Integrated Django Rest Framework (DRF) for building APIs.
- **Frontend Development:**
 - Explored and modified the SWFrontend framework for frontend development.
 - Enhanced user interface (UI) and usability of frontend components.
- **Database Management:**
 - Conducted meticulous database management tasks, including data entry corrections and validation.
 - Ensured data integrity and consistency within the database.
- **Document Generation:**
 - Generated PDF and Excel documents from application data.
 - Focused on design, formatting, and clarity of generated documents.
- **API Development:**
 - Developed a common API for communication between different parts of the software.
 - Conducted thorough testing and documentation of the API.
- **Laytime Calculator Optimization:**
 - Analyzed and improved the efficiency and accuracy of the laytime calculator.
 - Enhanced user interface for ease of use and understanding.
- **Workflow Management:**
 - Optimized workflow processes, including form saving sequences and approval initiation.
 - Implemented email notifications for various stages of the approval process.

- **UI Refinements:**

- Enhanced user interface elements, including form headers and BRM material tables.
- Improved button functionality and appearance for better user experience.

- **Dashboard Development:**

- Developed and refined the dashboard feature with data visualization elements.
- Improved overall usability and layout based on feedback and testing.

- **Technologies Learned:**

- Reactjs, Ant design, and Apache Echart for frontend development.
- Django and DRF for backend development and API creation.
- PostgreSQL for database management.
- SWFrontend framework for frontend modification and integration.

Snapshots of Training Activities:

➤ **CRUD Application**

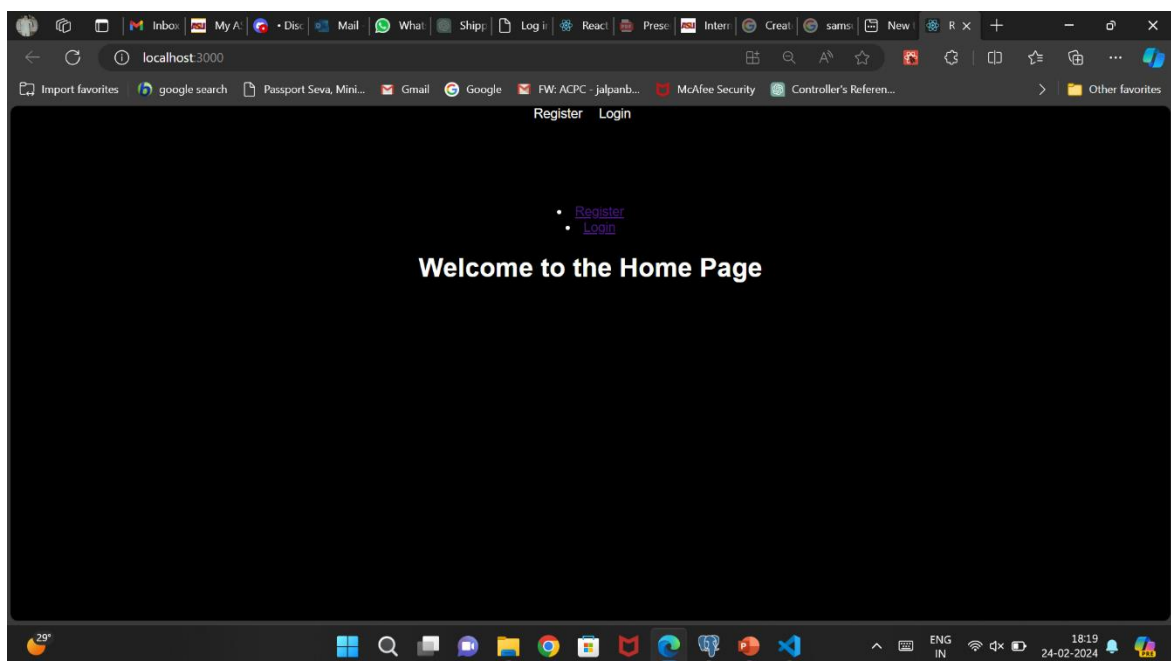


Figure 2.1 Home Page

- This figure shows the layout of the home page of CRUD application, including navigation elements and key features.

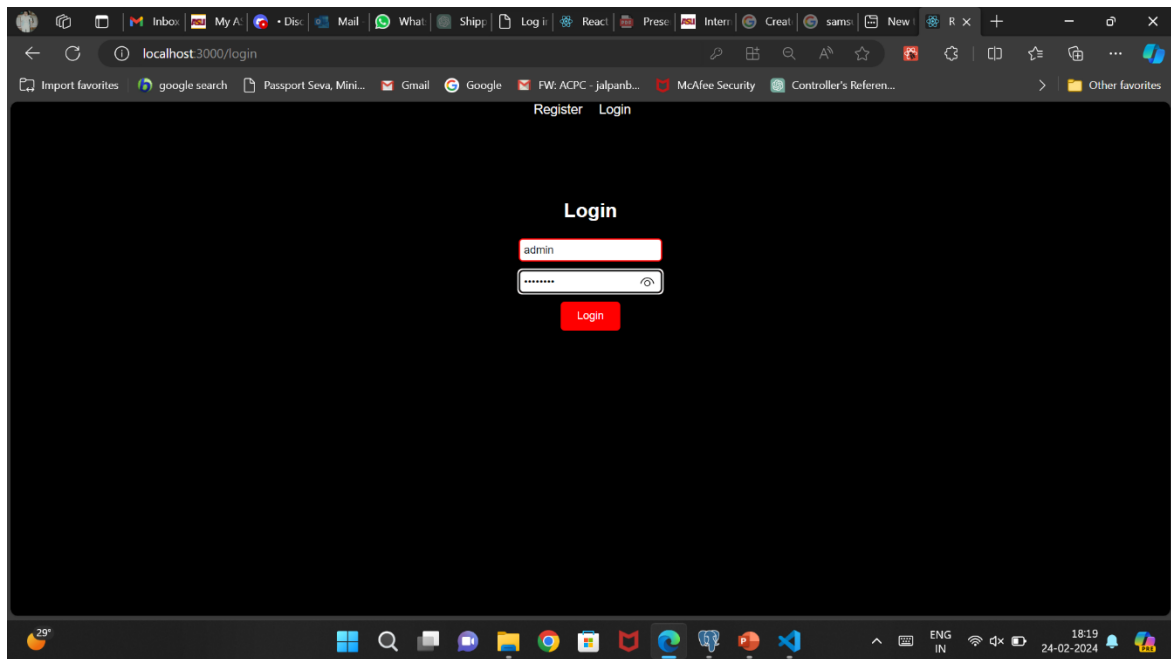


Figure 2.2 Login Page

- This figure displays the login interface of CRUD application, where users enter their credentials to access the system.

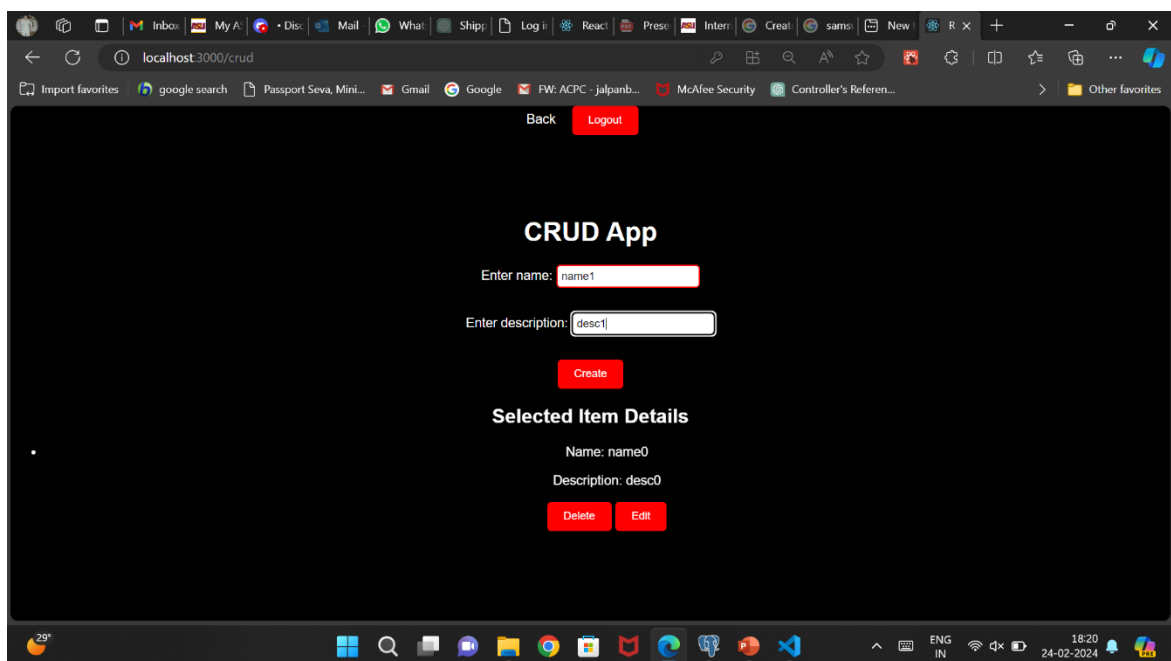


Figure 2.3 Entering a record

- This figure illustrates the process of entering a new record or data into the system.

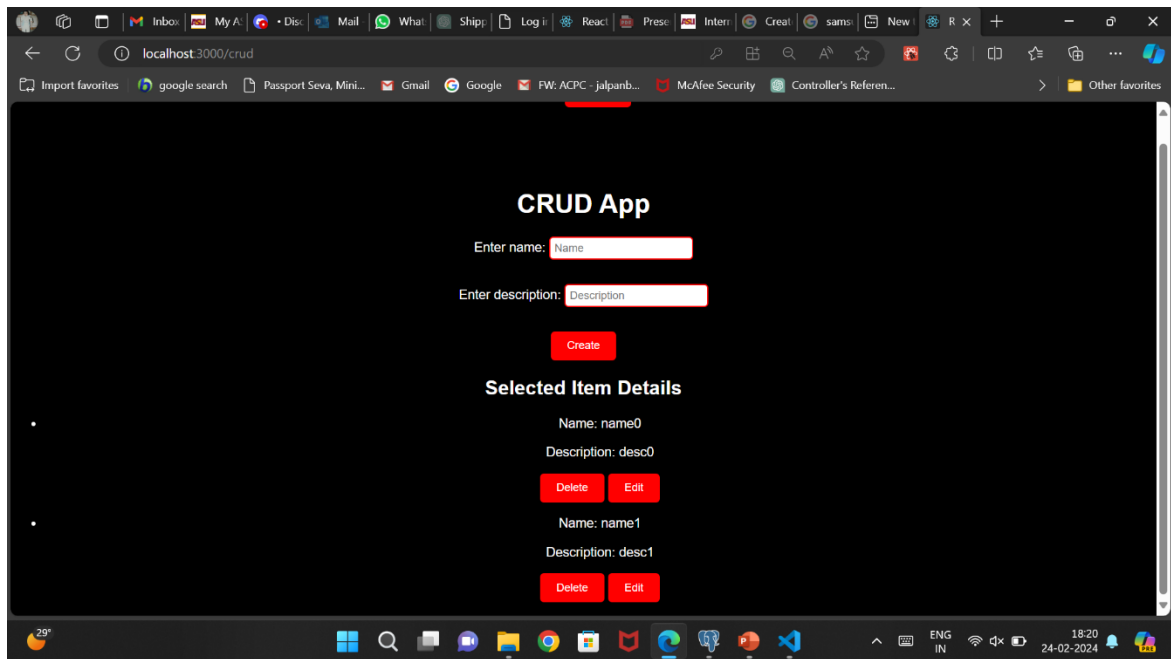


Figure 2.4 Record Created

- This figure confirms the successful creation of a record in the application.

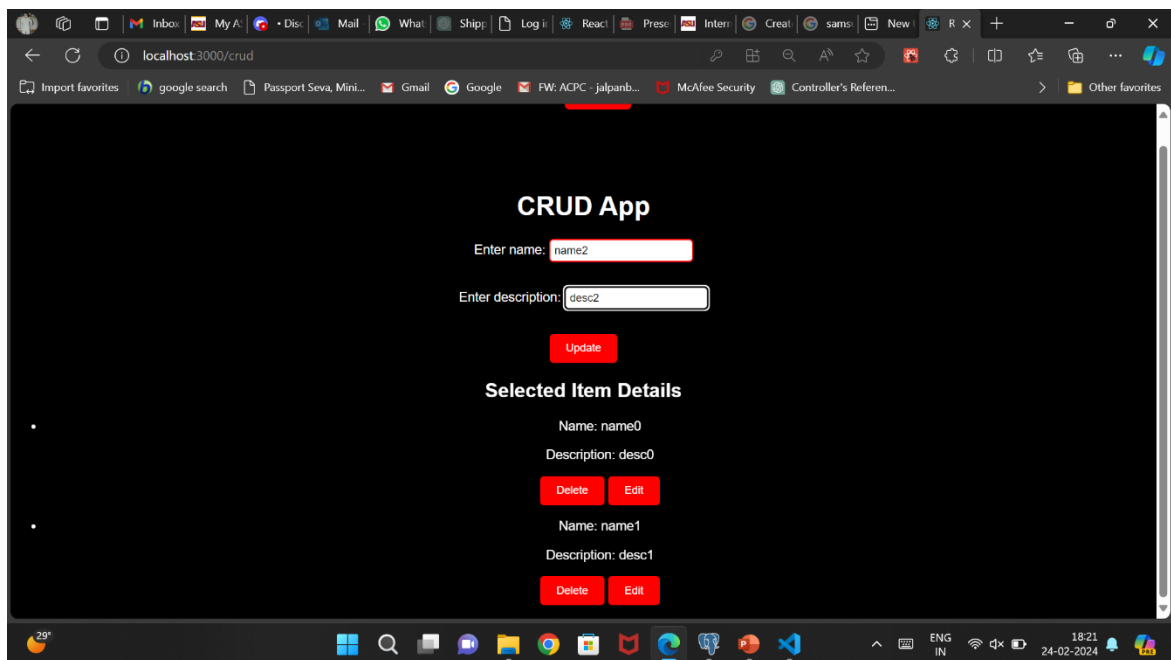


Figure 2.5 Updating a record

- This figure demonstrates how users can update an existing record or modify data within the system.

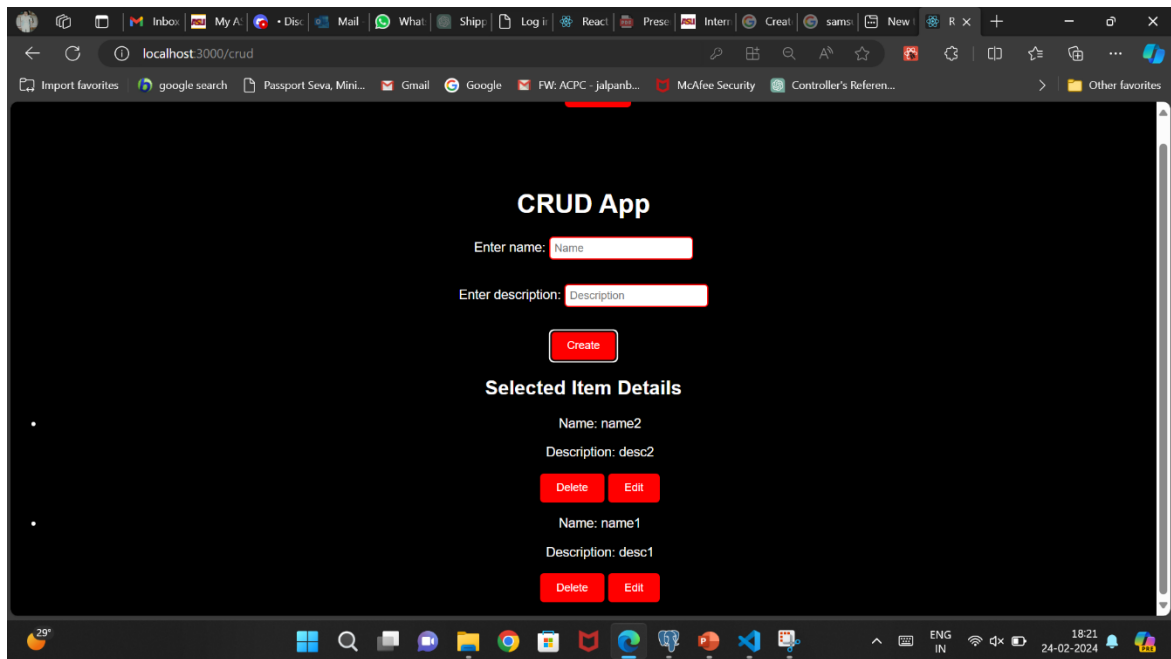


Figure 2.6 Updated the record

- This figure confirms the successful update of a record in the application.

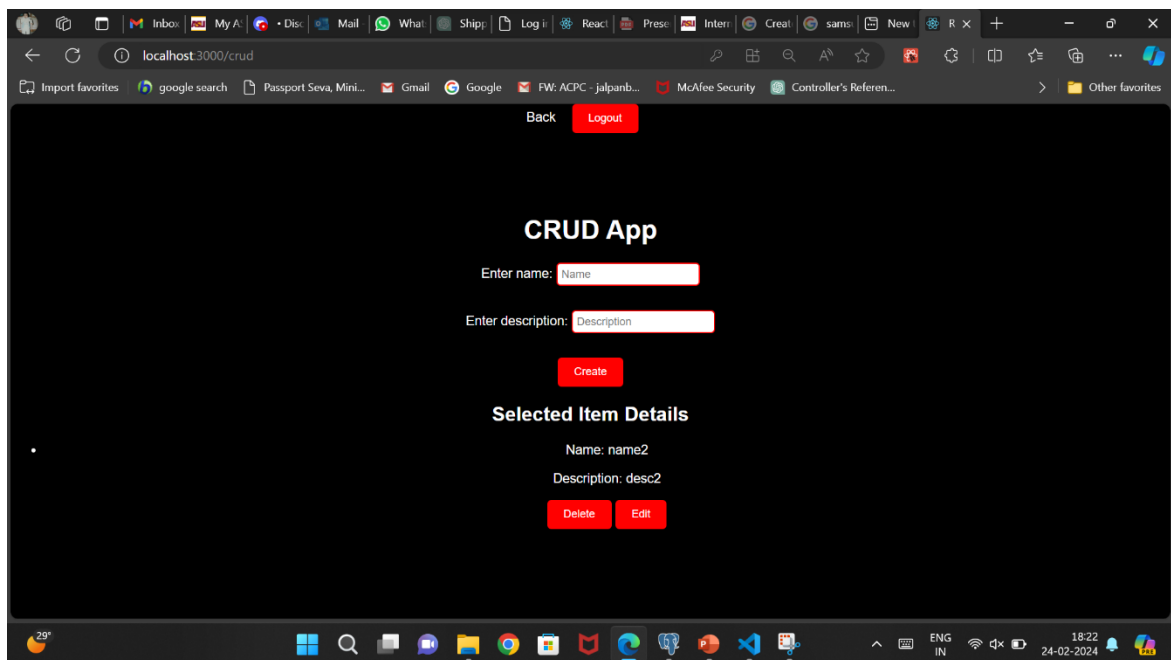


Figure 2.7 Deleted a record

- This figure shows the process of deleting a record from the system.

➤ Laytime Calculator Application

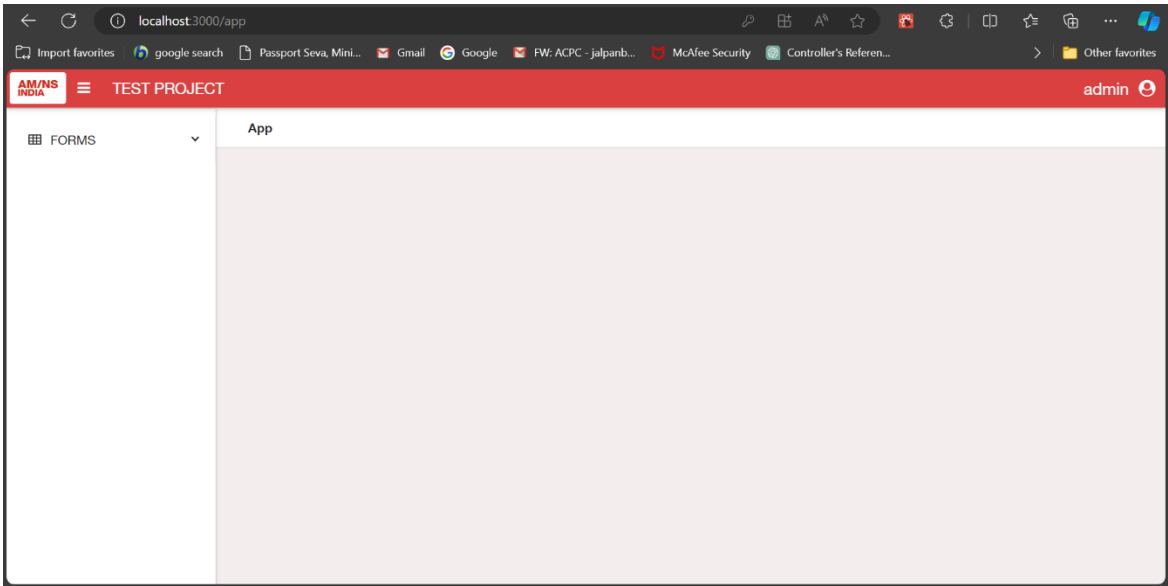


Figure 2.8 Created a subsection

- This figure indicates the creation of a section within the Laytime Calculator Website. Here the ‘FORMS’ section will contain subsection with a list view.

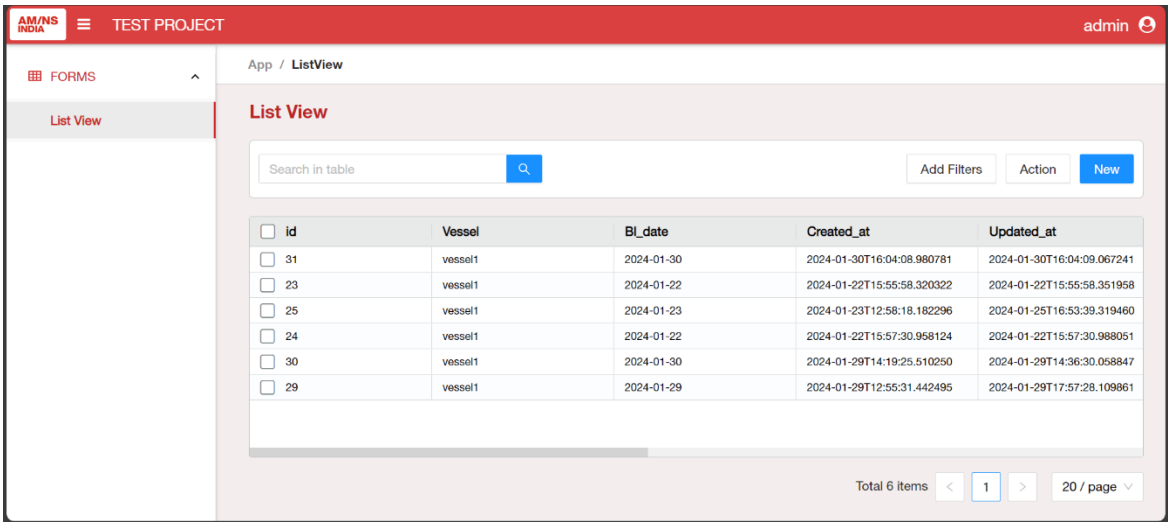


Figure 2.9 Adding a List View

- This figure shows the list view or displaying a list of items in the subsection in the Laytime Calculator Website.

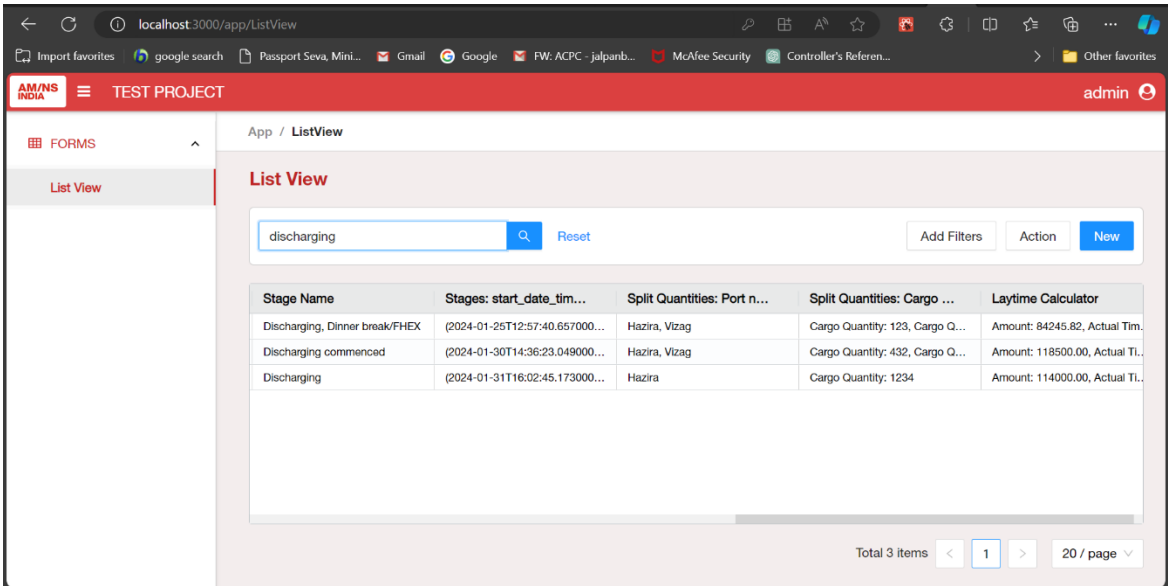


Figure 2.10 Testing generic search

- This figure displays the testing process for a generic search feature in the Laytime Calculator Website. After searching for ‘discharging’, the records containing ‘discharging’ in any column will be displayed on the list view.

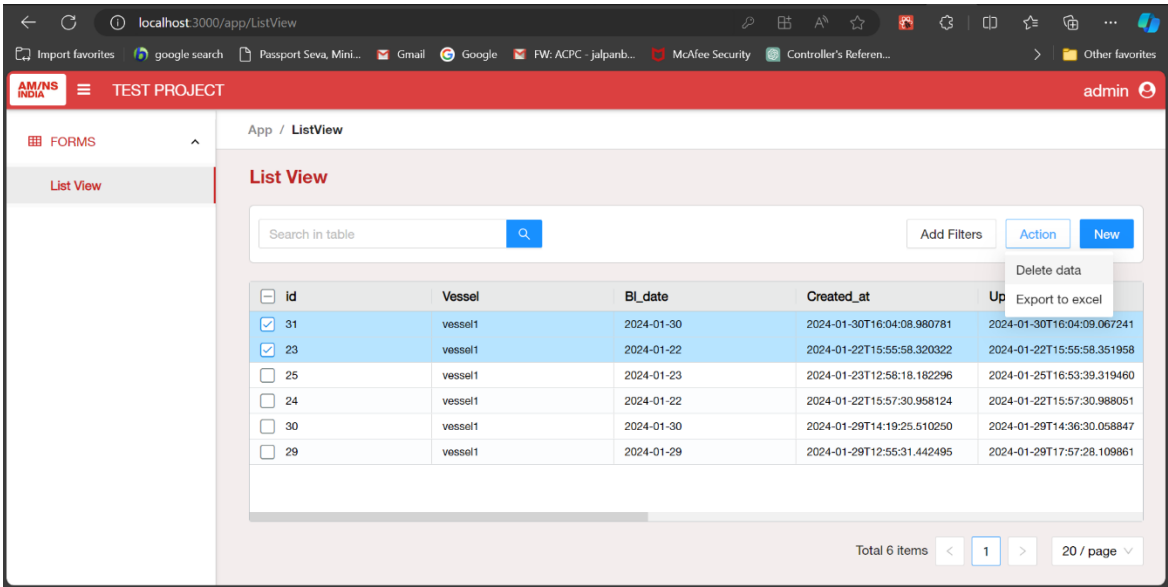


Figure 2.11 Actions to perform on records

- The figure shows the actions that can be performed on records within the Laytime Calculator Website.

The screenshot shows a web application interface for 'AM/NS INDIA' with a 'TEST PROJECT' header. The user is logged in as 'admin'. The left sidebar shows 'FORMS' and 'List View'. The main content area is titled 'List View' and contains a form for creating a new form. The form has the following fields:

- Vessel: vessel1
- B/L Date: 30-01-2024 (with a calendar dropdown for Jan 2024)
- Demurrage rate/day (USD \$): 12000
- Despatch rate/day (USD \$): 6000
- Cargo Qty (M/T): 2000
- Discharge rate: 100

Buttons for 'Back', 'Clear', and 'Save' are present at the top of the form.

Figure 2.12 Creating a new Form

- The figure indicates the process of creating a new form or inputting data in the Laytime Calculator Website.

The screenshot shows the same web application interface. The form is now for adding a shipping stage. The fields are:

- shipper1
- b1
- NOR Tendered: 03-02-2024 20:07
- Commenced Loading Time: 01-02-2024 20:07
- Completed Loading Time: 10-01-2024 20:07

A 'Calculate' button is located below the form. Below the form is a section titled 'Shipping Stage details' with a table:

Count	Stage Name	Start Date Time	End Date Time
1	Customs cleared	02-01-2024 20:07	03-01-2024 20:07

Below the table is a button '+ Add Shipping stage'.

Figure 2.13 Adding Shipping stage

- This figure displays the addition of a shipping stage or step in the Laytime Calculator Website.

The screenshot shows a web application interface for calculating laytime. At the top, there's a navigation bar with 'AM/NS INDIA' and 'TEST PROJECT'. A sidebar on the left has 'FORMS' and 'List View'. The main area contains input fields for 'shipper1' and 'b1'. Below these are three date-time pickers: 'NOR Tendered' (03-02-2024 20:07), 'Commenced Loading Time' (01-02-2024 20:07), and 'Completed Loading Time' (10-01-2024 20:07). A blue 'Calculate' button is positioned below the pickers. Below the button is a table with four columns: 'Total Allowed time', 'Total Actual time', 'Time Saved', and 'Total Despatch Amount'. The table contains the following data:

Total Allowed time	Total Actual time	Time Saved	Total Despatch Amount
20	1	19	114000

Below the table is a section titled 'Shipping Stage details' with a table that has four columns: 'Count', 'Stage Name', 'Start Date Time', and 'End Date Time'.

Figure 2.14 Calculating Despatch or Demurrage

- The figure shows the calculation process for despatch or demurrage in the Laytime Calculator Website.

The screenshot shows the same web application interface as Figure 2.14, but with additional shipping stages added. The 'Calculate' button is now at the top right. The table below it shows updated values: 'Total Allowed time' is 20, 'Total Actual time' is 1.33333, 'Time Saved' is 18.66667, and 'Total Despatch Amount' is 112000.02. Below the table is the 'Shipping Stage details' section, which now contains two rows of data:

Count	Stage Name	Start Date Time	End Date Time
☒	Customs cleared	02-01-2024 20:07	03-01-2024 20:07
☒	Discharging	02-01-2024 23:08	03-01-2024 07:08

At the bottom of the 'Shipping Stage details' section is a button labeled '+ Add Shipping stage'.

Figure 2.15 Adding multiple Shipping Stages and recalculating

- This indicates the process of adding multiple shipping stages and recalculating in the Laytime Calculator Website.

The screenshot shows a web browser at localhost:3000/app/ListView/new. The page has a red header with 'AM/NS INDIA' and 'TEST PROJECT' on the left, and 'admin' on the right. A sidebar on the left contains 'FORMS' and 'List View'. The main content area is titled 'Split Cargo Quantity'. It contains two rows of input fields: the first row has 'Port' (Hazira), 'Cargo Quantity' (1234), and 'Amount' (69104.01234); the second row has 'Port' (Vizag), 'Cargo Quantity' (567), and 'Amount' (31752.00567). Below these is a 'Remaining Cargo Quantity' field with the value 199. At the bottom right, there are buttons for 'Download Pdf' and 'Mail Laytime Excel Sheet'.

Figure 2.16 Splitting the quantity over ports

- This figure displays the process of splitting the quantity of items over different ports in the Laytime Calculator Website.

The screenshot shows the same web browser at localhost:3000/app/ListView/new. The page now displays a 'List View' form. A green success message 'Data added successfully!' is shown at the top. The form has a 'Back' button and 'Clear' and 'Save' buttons. The form fields are: 'Vessel' (vessel1), 'Demurrage rate/day (USD \$)' (12000), 'B/L Date' (30-01-2024), 'Despatch rate/day (USD \$)' (6000), 'Load Port' (port1), 'Cargo Qty (M/T)' (2000), and 'Discharge Port' (port1), 'Discharge rate' (100).

Figure 2.17 Saving a record

- The figure indicates the successful saving of a record or data in the Laytime Calculator Website.

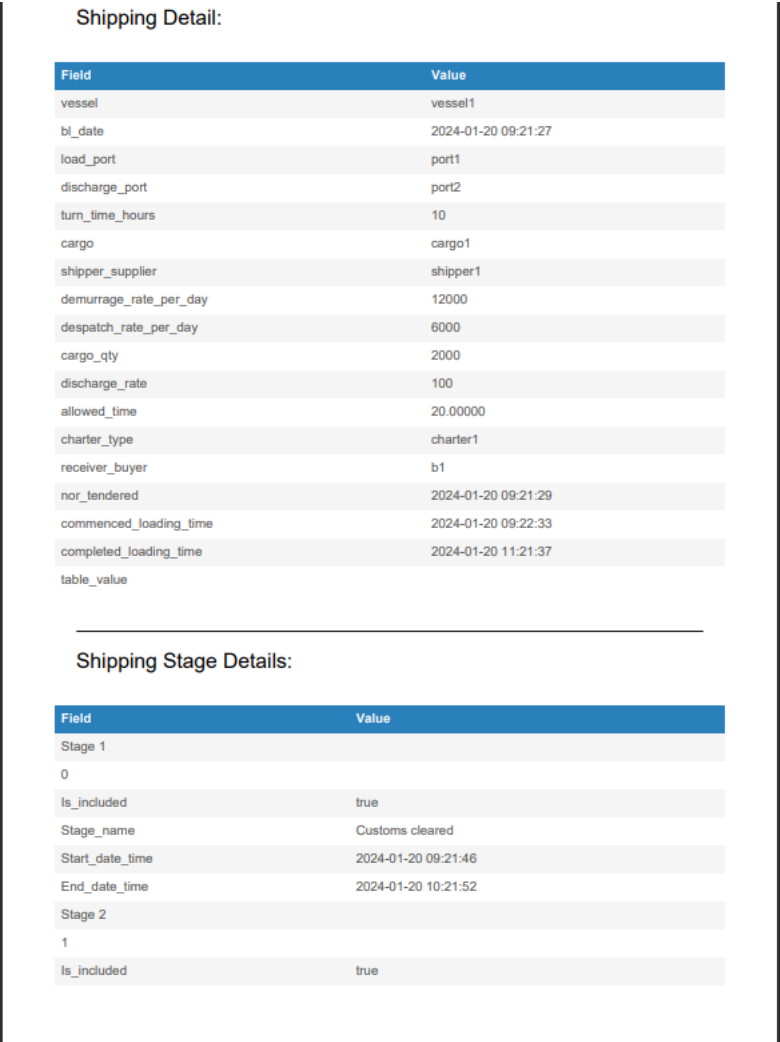


Figure 2.18 Using jspdf library for PDF creation Page-1

- The figure shows the usage of the jspdf library for creating PDFs in the Laytime Calculator Website. Page 1 of the PDF.



Figure 2.19 Using jspdf library for PDF creation Page-2

- The figure shows the usage of the jspdf library for creating PDFs in the Laytime Calculator Website. Page 2 of the PDF.

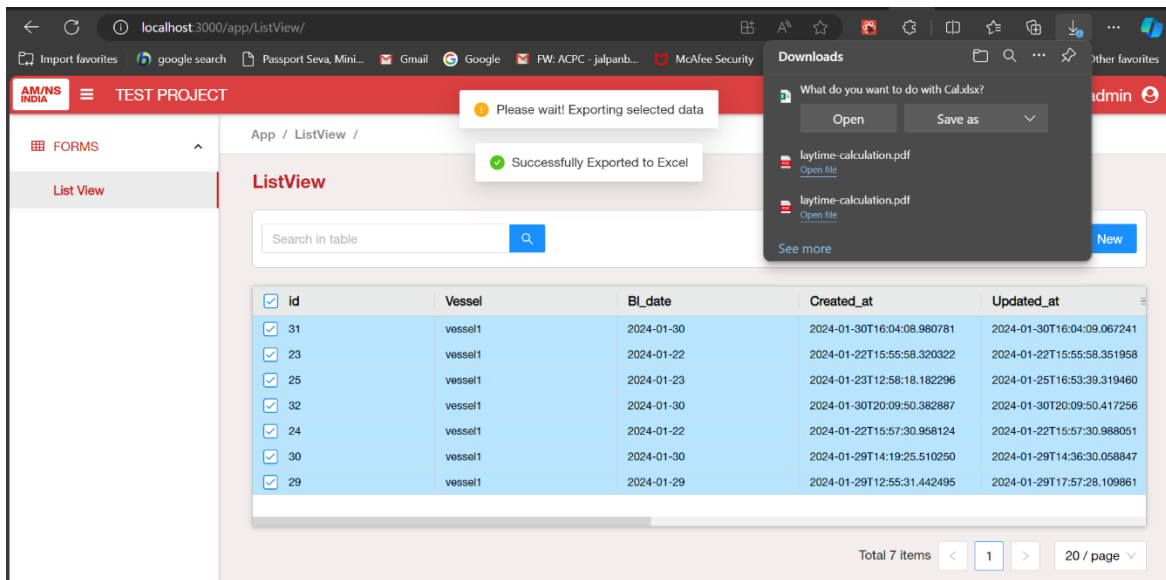


Figure 2.20 Selecting multiple record to export it in excel

This figure indicates the selection of multiple records for exporting them into an Excel file in the Laytime Calculator Website.

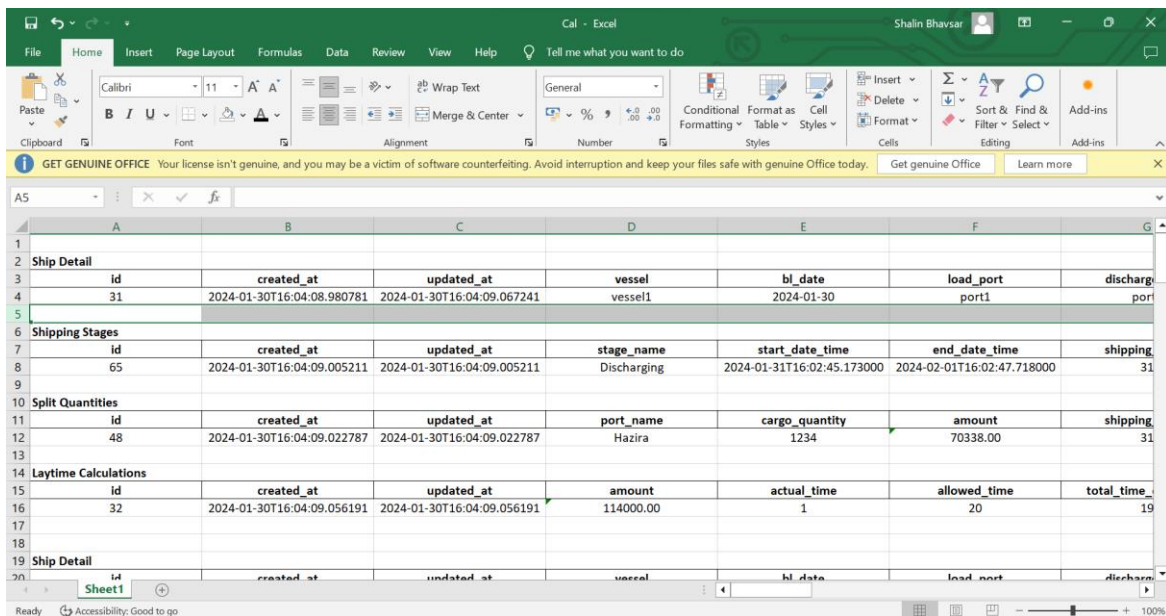


Figure 2.21 Excel file

➤ This figure shows the exported Excel file containing data from the Laytime Calculator Website.

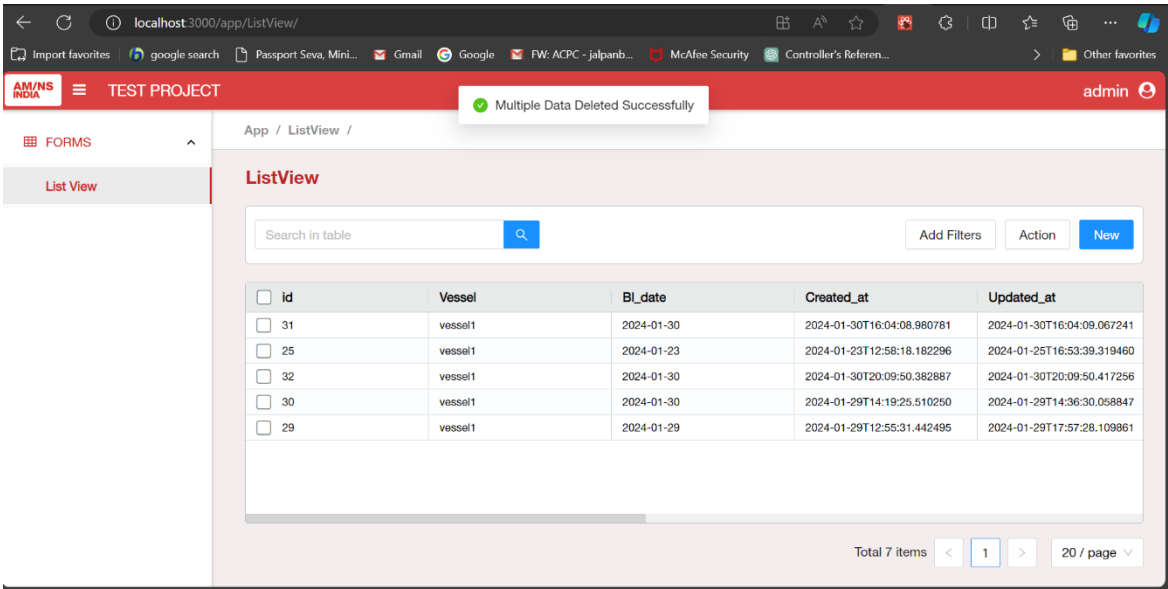


Figure 2.22 Deleting multiple record

- Here the figure illustrates the process of deleting multiple records simultaneously in the Laytime Calculator Website.

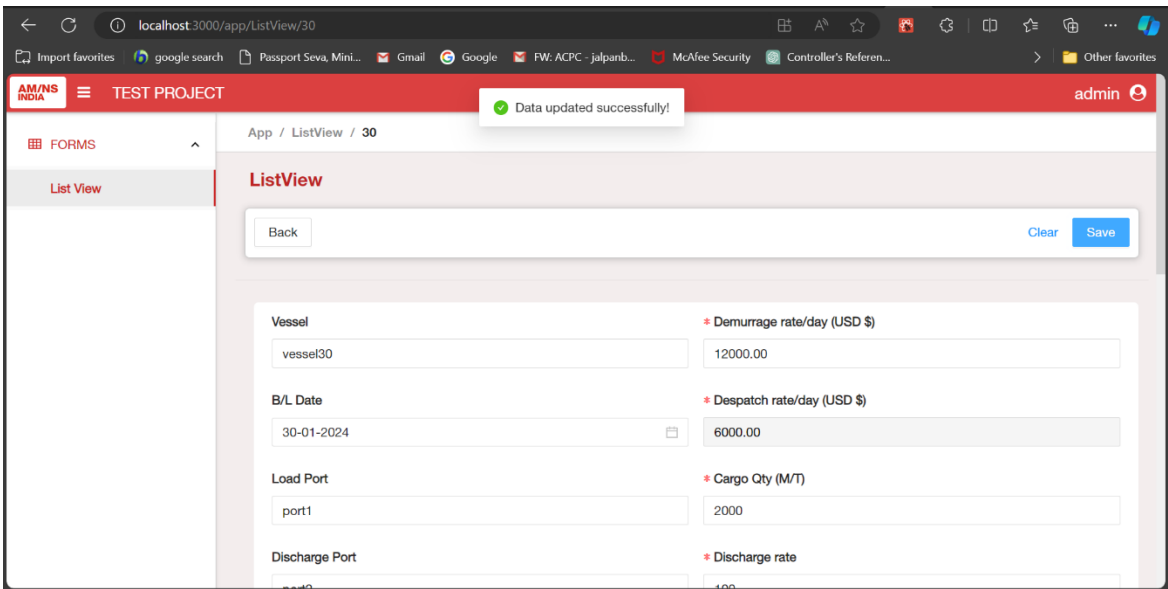


Figure 2.23 Updating a record

- The figure shows the interface or process for updating/editing an existing record in the Laytime Calculator Website. Here the name of the vessel is changed from vessel1 to vessel30.

Figure 2.24 Adding the Initiate Approval button in the Form Header

- Here the form Header is updated and instead of a Save button an Action button is added which will allow the user to initiate the approval process, and update the record by saving and using the download button to download PDF or Excel for the specific record.

Figure 2.25 Approval Form subsection after clicking the Initiate Approval button

- This figure shows the approval form after clicking on the Initiate Approval button where the approver can check the record, add a remark, and click on the Approve/Reject button to either Approve or Reject the record.

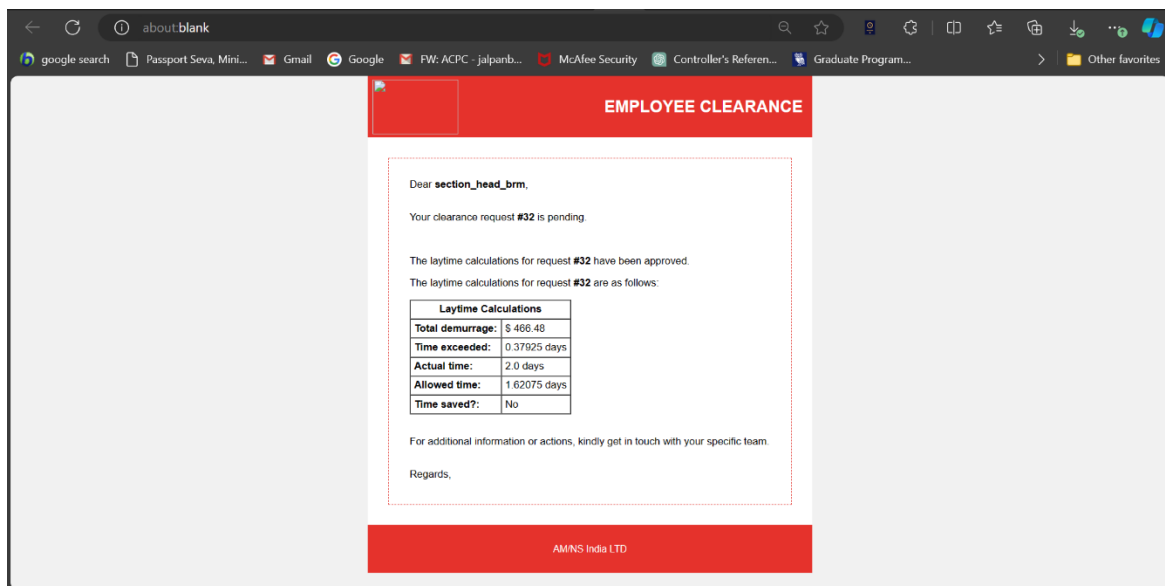


Figure 2.26 Email test case after approval

- Here, this figure shows a test email template that will appear after approval from approver. The email template contains the message to the next approver informing them that their action on the specific record is pending. Also, there is a laytime calculation table for their reference.

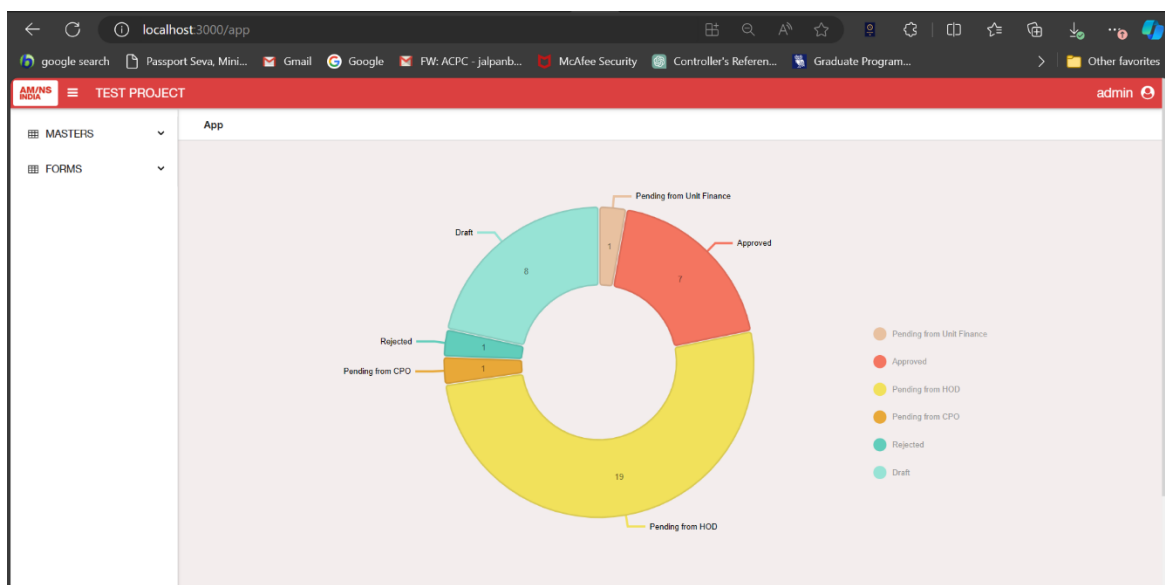


Figure 2.27 Dashboard to View the status of records

- This figure shows a dashboard that is created using 'nivo' charts. The dashboard shows the status of the records for the specific user.

CHAPTER 3 SYSTEM PLANNING

3.1 Project Development Approach

Each project needs to be developed with a software model which makes the project high-quality, reliable, and cost-effective.

- Name of the software model: Agile Software Development Model
- Agile is a software development approach where requirements and solutions evolve through collaboration between self-organizing cross-functional teams. It promotes adaptive planning, evolutionary development, early delivery, and continuous improvement, and it encourages rapid and flexible responses to change.
- Justification: Agile was chosen for this project due to its ability to handle changes in requirements, which is common in software development projects. It allows for iterative development, ensuring that the project remains aligned with the client's needs throughout the development process.
- Advantages:
 - Allows for flexibility and adaptability to changing requirements.
 - Promotes customer satisfaction through early and continuous delivery of valuable software.
 - Enhances collaboration and communication between team members and stakeholders.
 - Enables quick response to feedback and changes, reducing the risk of project failure.

3.2 System Modules

In the Laytime Calculator, the project is organized into distinct modules, each playing a crucial role in delivering a seamless and personalized user experience. This section provides a brief overview of the primary modules within the system.

3.2.1 *User Authentication and Authorization*

This module manages user authentication and authorization for accessing the Laytime Calculator Application. It ensures that only authorized users can access the application and perform actions based on their roles.

3.2.2 Data Entry and Management

This module handles the input and management of essential information related to vessel operations. It includes functionalities for entering vessel details, cargo specifics, NOR tendered date, commenced loading date, completed loading date, and other relevant information.

3.2.3 Dynamic Stage Addition

This module allows users to add stages with customizable names and date ranges to represent different segments of the vessel operation. It provides the flexibility to accommodate varying stages in vessel operations.

3.2.4 Real-Time Laytime Calculation

This module performs real-time calculations to determine the duration of each stage and the total laytime for the entire operation. It compares the total laytime with the allowed time to identify whether there is extra time (despatch) or exceeded time (demurrage).

3.2.5 Database Integration

This module integrates with a relational database management system (RDBMS) to store laytime calculation data securely. It stores information related to vessel details, cargo specifics, stagewise data, laytime results, and other relevant parameters.

3.2.6 Data Export and Reporting

This module allows users to export laytime calculation data, including stagewise details and results, to an Excel file for recordkeeping and further analysis. It also enables users to generate Demurrage reports by querying the database based on specific criteria such as vessel name, port, charter type, or dates.

3.2.7 Business Intelligence Support

This module utilizes the integrated database as a repository for vessel-related data, enabling various departments, including bulk raw material, shipping, and chartering, to conduct business intelligence. It allows for trend analysis, performance evaluation, and informed decision-making based on historical laytime records.

3.2.8 Approval Hierarchy and Payment Trail

This module implements an approval hierarchy for laytime calculations. It includes functionalities for defining the approval flow, specifying delegation rules, and tracking the

approval status of laytime calculations. The module ensures compliance with regulations and contractual agreements by enforcing the approval hierarchy.

3.3 Functional Requirements

ID	Title and Description
FR1	<i>Title:</i> User Authentication and Authorization
	<i>Description:</i> The system should authenticate users and manage their authorization levels to access the Laytime Calculator Application.
FR2	<i>Title:</i> Data Entry and Management
	<i>Description:</i> Users should be able to enter and manage essential information related to vessel operations, such as vessel details, cargo specifics, NOR tendered date, commenced loading date, and completed loading date.
FR3	<i>Title:</i> Dynamic Stage Addition
	<i>Description:</i> Users should be able to add stages with customizable names and date ranges to represent different segments of the vessel operation.
FR4	<i>Title:</i> Real-Time Laytime Calculation
	<i>Description:</i> The system should perform real-time calculations to determine the duration of each stage and the total laytime for the entire operation. It should also compare the total laytime with the allowed time to identify extra time (despatch) or exceeded time (demurrage).
FR5	<i>Title:</i> Database Integration
	<i>Description:</i> The system should integrate with a relational database management system (RDBMS) to securely store laytime calculation data, including vessel details, cargo specifics, stagewise data, laytime results, and other relevant parameters.
FR6	<i>Title:</i> Data Export and Reporting
	<i>Description:</i> Users should be able to export laytime calculation data, including stagewise details and results, to an Excel file for recordkeeping and further analysis. It should also allow users to generate Demurrage reports by querying the database based on specific criteria such as vessel name, port, charter type, or

	dates.
FR7	<i>Title:</i> Business Intelligence Support
	<i>Description:</i> The system should serve as a repository for vessel-related data, enabling various departments to conduct business intelligence. It should allow for trend analysis, performance evaluation, and decision-making based on historical laytime records.
FR8	<i>Title:</i> Approval Hierarchy
	<i>Description:</i> The system should implement an approval hierarchy for laytime calculations. It should include functionalities for defining the approval flow, specifying delegation rules, and tracking the approval status of laytime calculations.

3.4 Non Functional Requirements

ID	Title and Description
NFR1	<i>Title:</i> Performance
	<i>Description:</i> The system shall be able to handle a large volume of users and data without significant performance degradation. Response times for calculations and data retrieval shall be within acceptable limits.
NFR2	<i>Title:</i> Security
	<i>Description:</i> The system shall implement robust security measures to protect sensitive data. This includes user authentication, data encryption, and access control mechanisms.
NFR3	<i>Title:</i> Reliability
	<i>Description:</i> The system shall be reliable, with minimal downtime and high availability. It shall be able to recover from failures gracefully and maintain data integrity.
NFR4	<i>Title:</i> Usability

	<i>Description:</i> The system shall have a user-friendly interface that is easy to navigate and understand. It shall provide clear instructions and feedback to users.
NFR5	<i>Title:</i> Scalability
	<i>Description:</i> The system shall be scalable, allowing for easy expansion to accommodate growing user and data requirements. It shall be able to handle increases in workload without significant changes to the underlying architecture.
NFR6	<i>Title:</i> Compatibility
	<i>Description:</i> The system shall be compatible with a wide range of devices and browsers. It shall support common operating systems and ensure consistency in functionality and appearance across different platforms.
NFR7	<i>Title:</i> Maintainability
	<i>Description:</i> The system shall be easy to maintain and update. It shall have well-organized code, documentation, and support for version control to facilitate future modifications.
NFR8	<i>Title:</i> Compliance
	<i>Description:</i> The system shall comply with relevant regulations and standards for data protection, privacy, and security. It shall also adhere to industry best practices for software development and deployment.

3.5 Timeline Chart

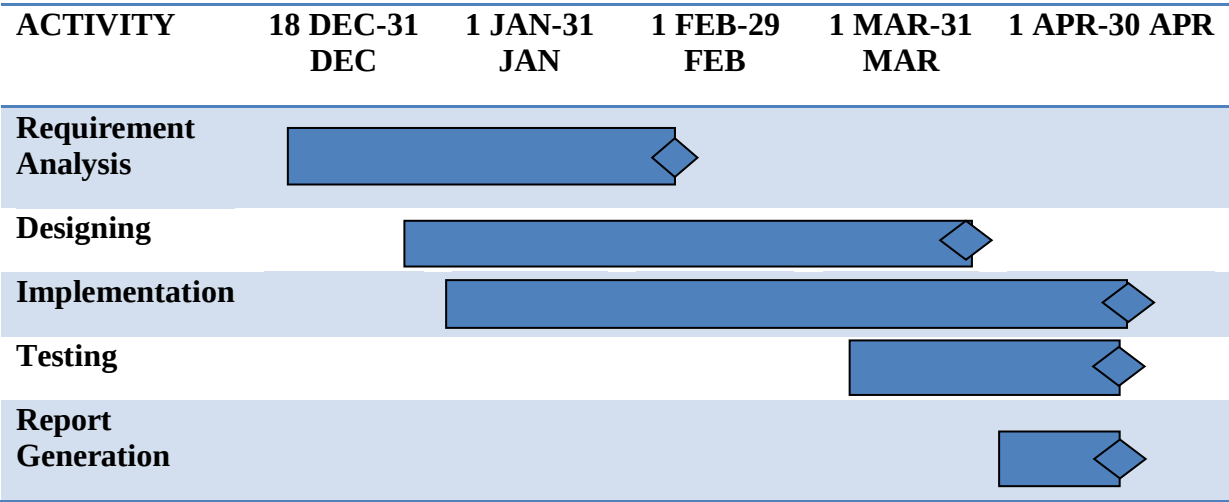


Figure 3.1 Timeline Chart

CHAPTER 4 SYSTEM DESIGN

4.1 Use Case Diagram

The Use case diagram provides a visual representation of the use cases within the Laytime Calculator system, highlighting user interactions and system functions.



Figure 4.1 Use case Diagram

4.2 Sequence Diagram

- **For User:** This figure depicts a sequential diagram specifically for user interactions, showing the flow of actions or events.

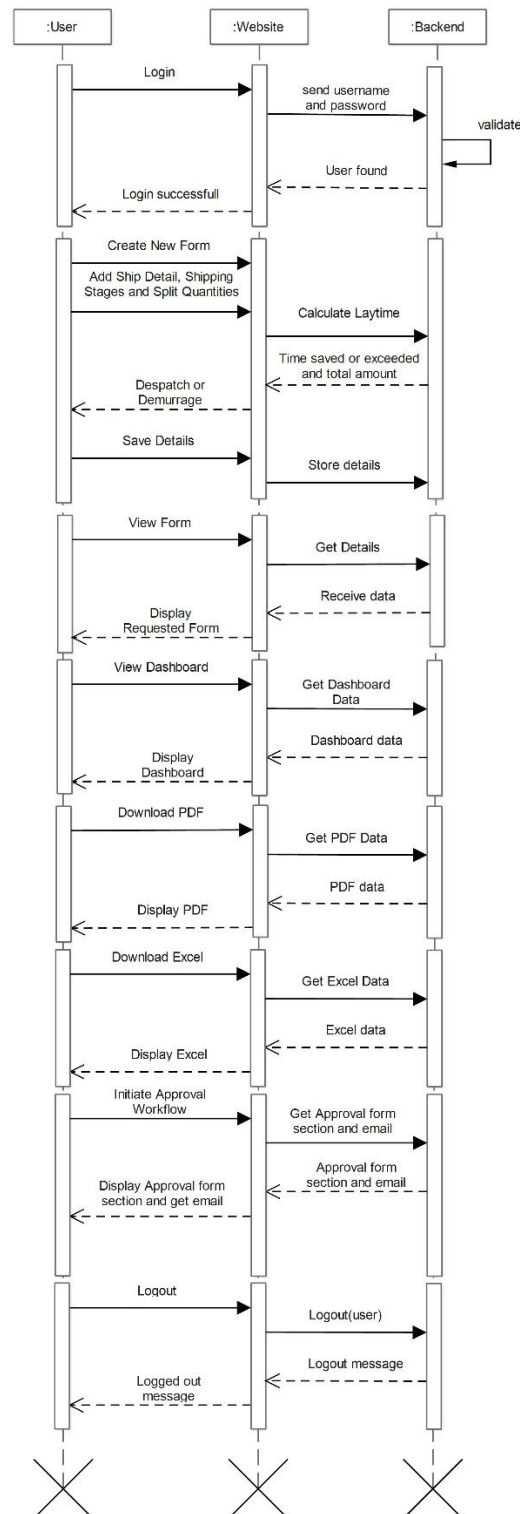


Figure 4.2 Sequential Diagram (For User)

- **For Approver:** This figure is a sequential diagram focused on the approval process, outlining the steps taken by approvers.

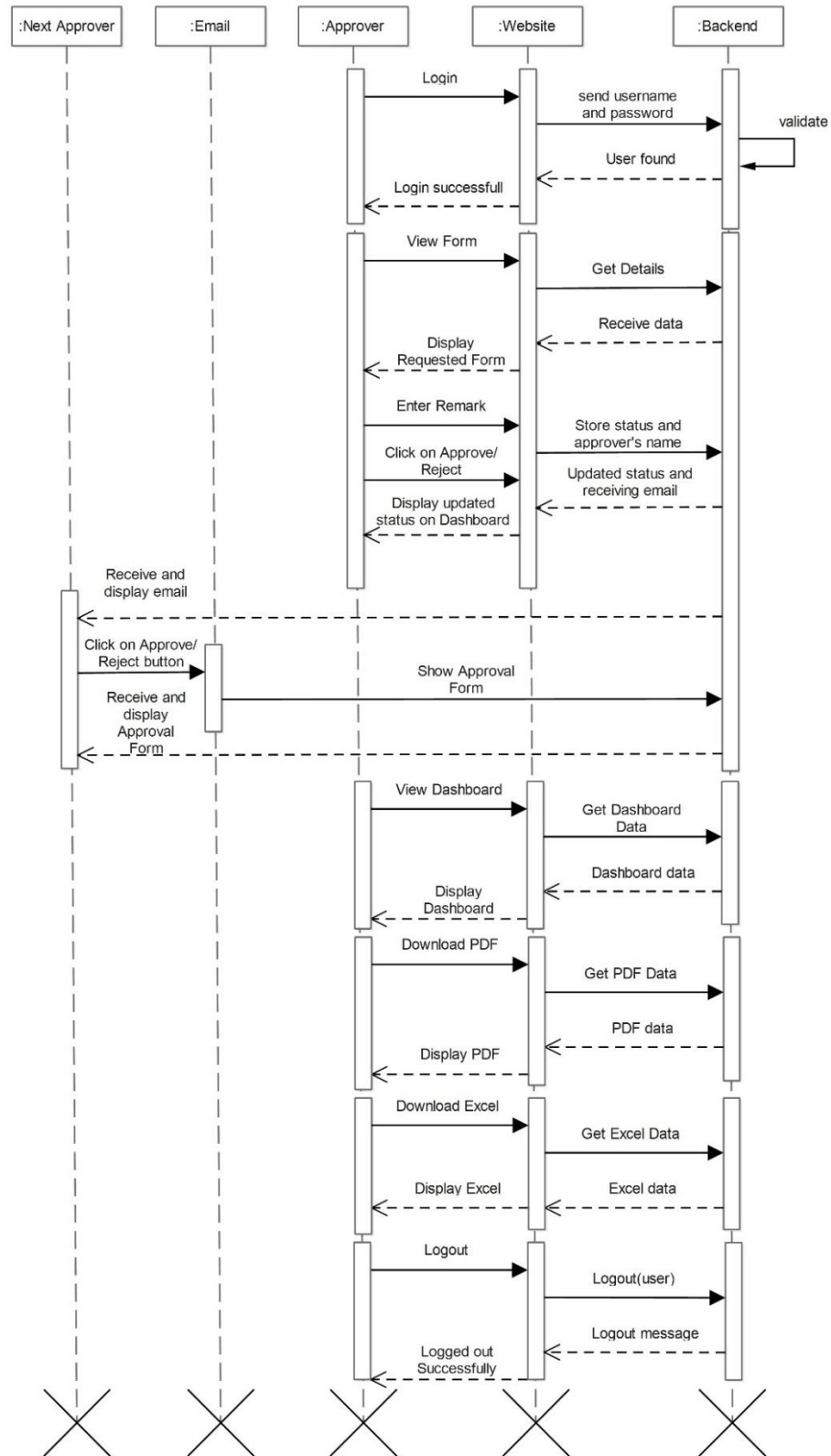


Figure 4.3 Sequential Diagram (For Approver)

4.3 Activity Diagram

This figure presents an activity diagram, mapping out the activities and processes within your system.

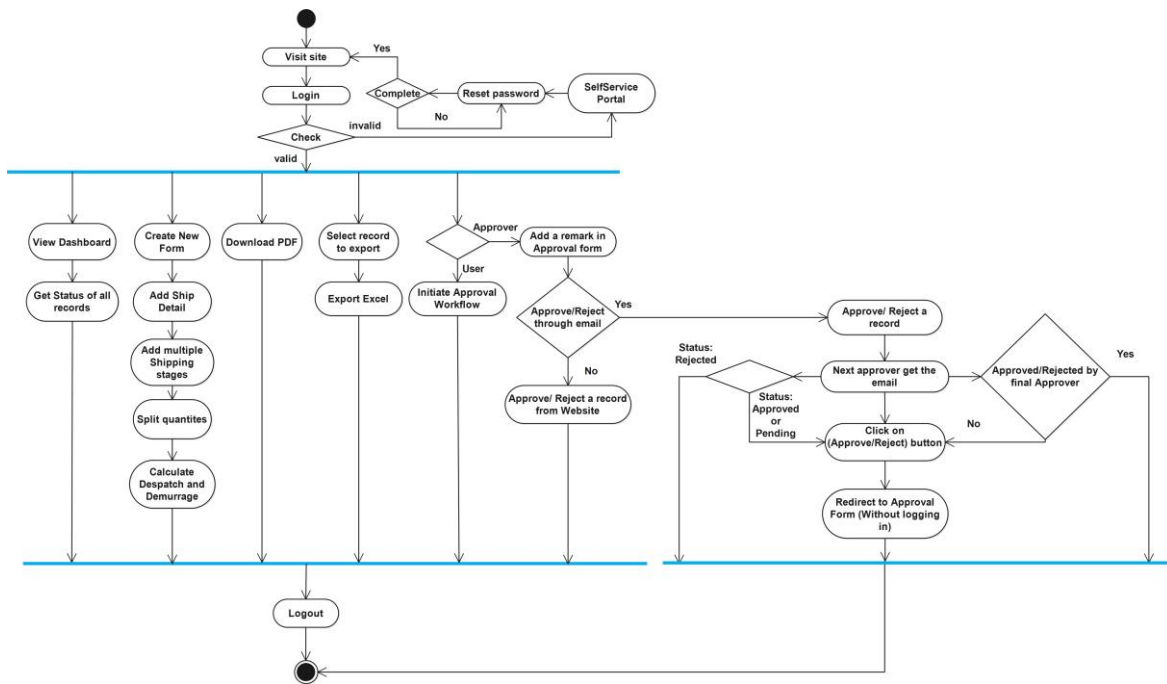


Figure 4.4 Activity Diagram

4.4 Class Diagram

This figure is a class diagram illustrating the structure and relationships of classes or objects in your system.

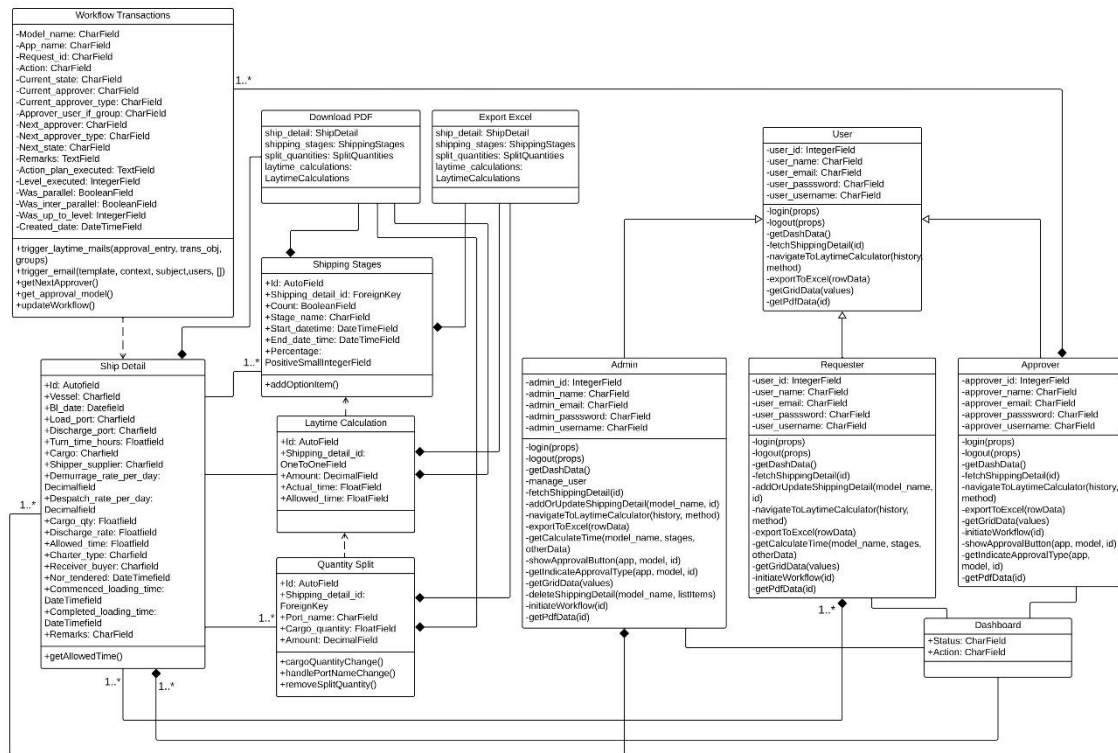


Figure 4.5 Class Diagram

4.5 Database Schema

Table 4.1 Ship Detail Table

Column Name	Data Type	Size	Constraint	Description
Id	Autofield	-	Primary Key	Unique identifier for each record
Vessel	Charfield	255	Null=True, Blank=True	Vessel name
Bl_Date	Datefield	-	Null=True, Blank=True	Bill of lading date
Load_port	Charfield	255	Null=True, Blank=True	Load port
Discharge_port	Charfield	255	Null=True, Blank=True	Discharge port
Turn_time_hours	Floatfield	-	-	Turn time in hours
Cargo	Charfield	255	Null=True, Blank=True	Cargo details
Shipper_supplier	Charfield	255	Null=True, Blank=True	Shipper or supplier details
Demurrage_rate_per_day	Decimalfield	10,2	-	Demurrage rate per day
Despatch_rate_per_day	Decimalfield	10,2	-	Despatch rate per day
Cargo_qty	Floatfield	-	-	Cargo quantity
Discharge_rate	Floatfield	-	-	Discharge rate

Allowed_time	Floatfield	-	-	Allowed time for operation
Charter_type	Charfield	255	Null=True, Blank=True	Charter type
Receiver_buyer	Charfield	255	Null=True, Blank=True	Receiver or buyer details
Nor_tendered	DateTimeField	-	-	NOR tendered date and time
Commenced_loading_time	DateTimeField	-	-	Commenced loading date and time
Completed_loading_time	DateTimeField	-	-	Completed loading date and time
Remarks	CharField	2000	Null=True, Blank=True	Remarks

Table 4.2 Shipping Stages Table

Column Name	Data Type	Size	Constraint	Description
Id	AutoField	-	Primary Key	Unique identifier for each record
Shipping_detail_id	ForeignKey	-	On Delete CASCADE	Foreign key reference to ShippingDetail
Count	BooleanField	-	-	Boolean value
Stage_name	CharField	255	-	Name of the stage

Start_datetime	DateTimeField	-	-	Start date and time of the stage
End_date_time	DateTimeField	-	-	End date and time of the stage
Percentage	PositiveSmallIntegerField	-	MinValueValidator(0), MaxValueValidator(100)	Percentage value (default 100)

Table 4.3 Quantity Split Table

ColumnName	Data Type	Size	Constraint	Description
Id	AutoField	-	Primary Key	Unique identifier for each record
Shipping_detail_id	ForeignKey	-	On Delete CASCADE	Foreign key reference to ShippingDetail
Port_name	CharField	255	-	Name of the port
Cargo_quantity	FloatField	-	-	Quantity of cargo
Amount	DecimalField	10,2	-	Amount

Table 4.4 Laytime Calculation Table

ColumnName	Data Type	Size	Constraint	Description
Id	AutoField	-	Primary Key	Unique identifier for

				each record
Shipping_detail_id	OneToOneField	-	On Delete CASCADE	One-to-one relationship with ShippingDetail
Amount	DecimalField	10,2	-	Amount
Actual_time	FloatField	-	Default=170382 3476.82923, Null=True, Blank=True	Actual time
Allowed_time	FloatField	-	-	Allowed time

Table 4.5 Workflow Table

ColumnName	Data Type	Size	Constraint	Description
Model_name	CharField	50	-	Model name of the object
App_name	CharField	50	-	Name of the associated application
Request_id	CharField	50	-	Unique identifier for the request
Action	CharField	50	blank=True, null=True, default="Initiate d"	Action taken on the request
Current_state	CharField	50	default="None",	Current state of

			null=True, blank=True	the request
Current_approver	CharField	3000	null=True, blank=True	Current approver of the request
Current_approver_type	CharField	30	null=True, blank=True	Type of the current approver
Approver_user_if_group	CharField	30	null=True, blank=True	User type if the approver is a group
Next_approver	CharField	3000	null=True, blank=True	Next approver in the workflow
Next_approver_type	CharField	30	null=True, blank=True	Type of the next approver
Next_state	CharField	50	default="Draft"	State to which the request will move next
Remarks	TextField	250	null=True, blank=True	Additional comments or notes
Action_plan_executed	TextField	-	null=True, blank=True	Details of the executed action plan
Level_executed	IntegerField	-	null=True, blank=True	Level at which the action was executed

Was_parallel	BooleanField	-	default=False	Indicates if the action was parallel
Was_inter_parallel	BooleanField	-	default=False	Indicates if the action was inter-parallel
Was_up_to_level	IntegerField	-	null=True, blank=True	Level up to which the action was executed
Created_date	DateTimeField	-	auto_now=True	Date and time when the object was created

4.6 Data Flow Diagram

This figure is a Level 0 Data Flow Diagram (DFD), providing an overview of the major processes and interactions in the system.

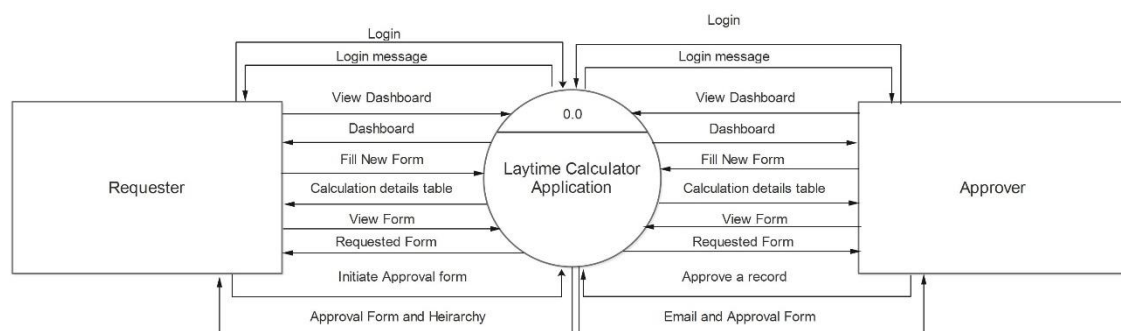


Figure 4.6 DFD Level 0

This figure is a Level 1 Data Flow Diagram (DFD), delving deeper into the processes and data flows identified in the Level 0 DFD.

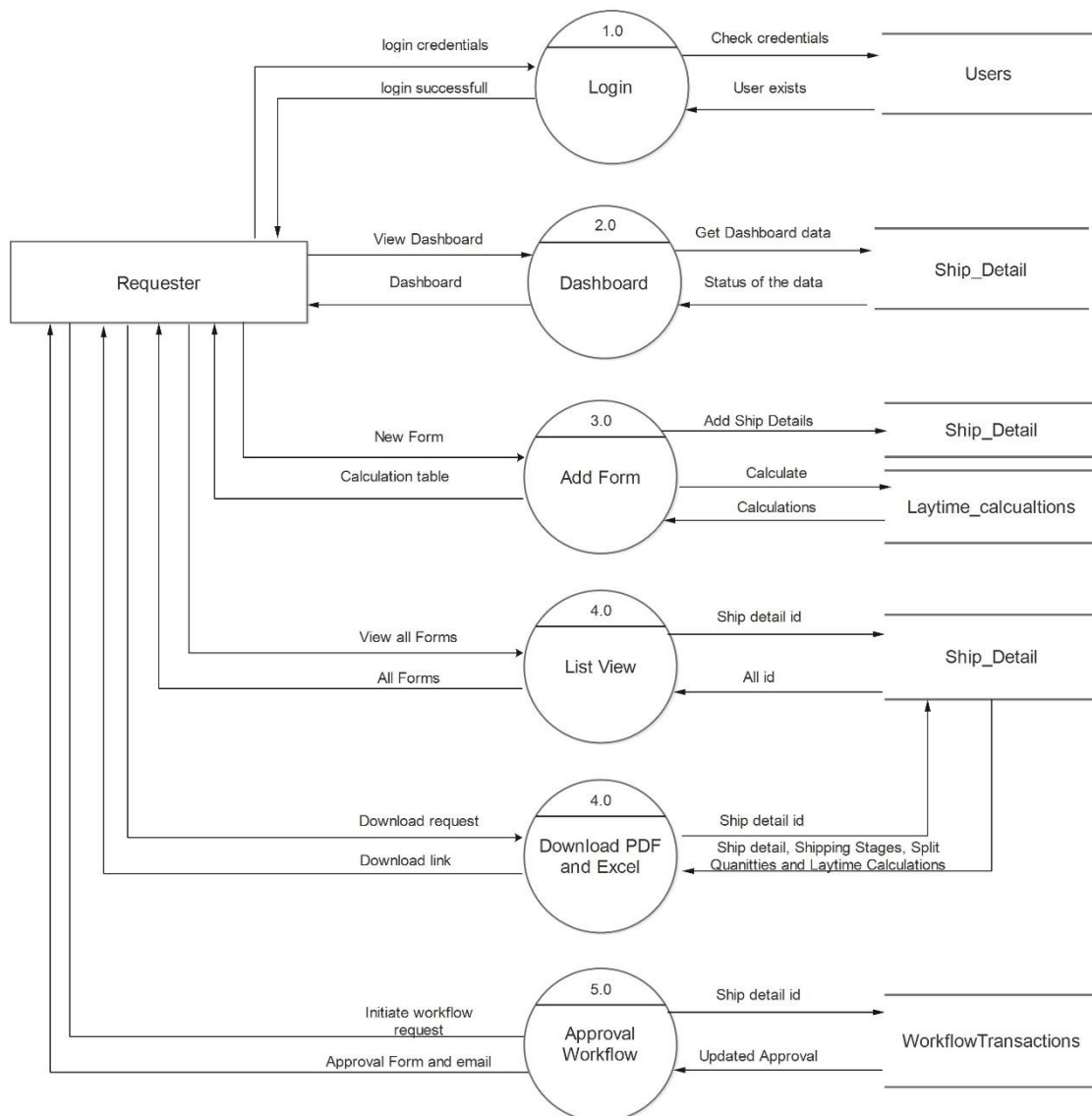


Figure 4.7 DFD Level 1

This figure represents a Level 2 Data Flow Diagram (DFD) for a adding a new form.

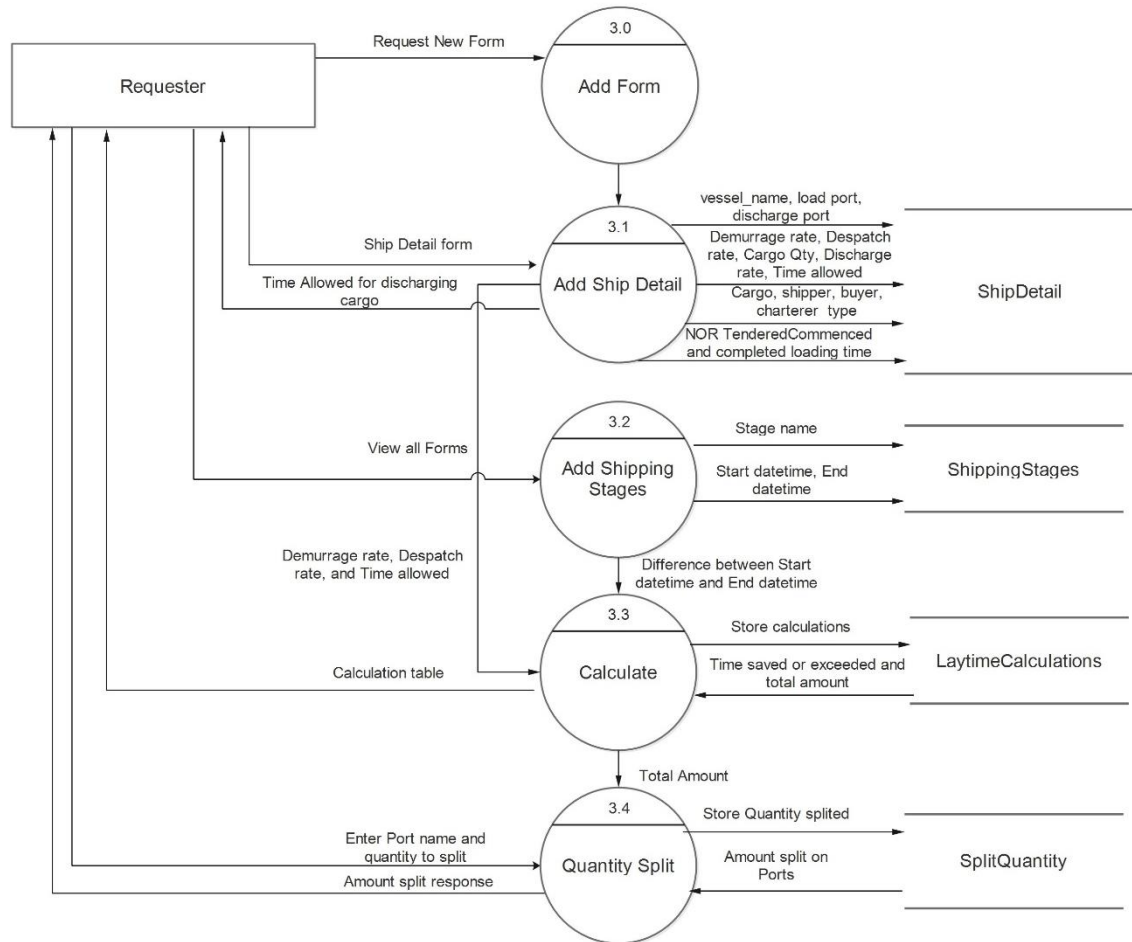


Figure 4.8 DFD Level 2 (3.0)

This figure represents another Level 2 Data Flow Diagram (DFD) for Approving a record by approver.

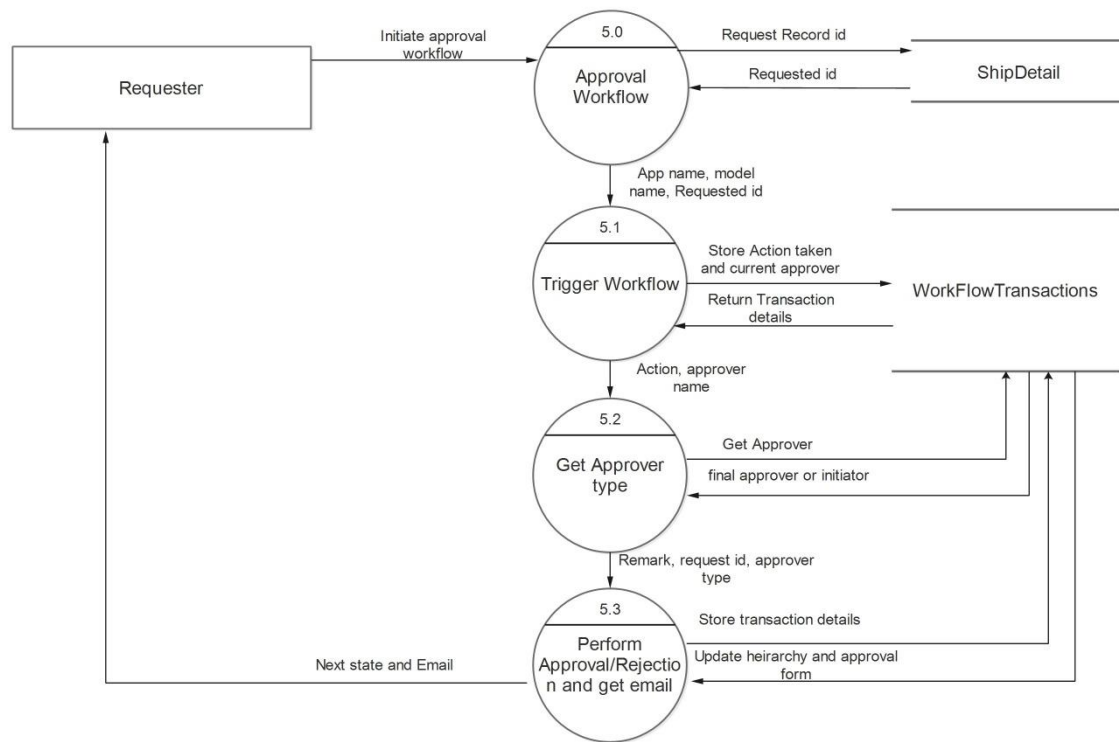


Figure 4.9 DFD Level 2 (5.0)

4.7 ER Diagram

This figure is an Entity-Relationship (ER) diagram, illustrating the entities and their relationships in your database schema.

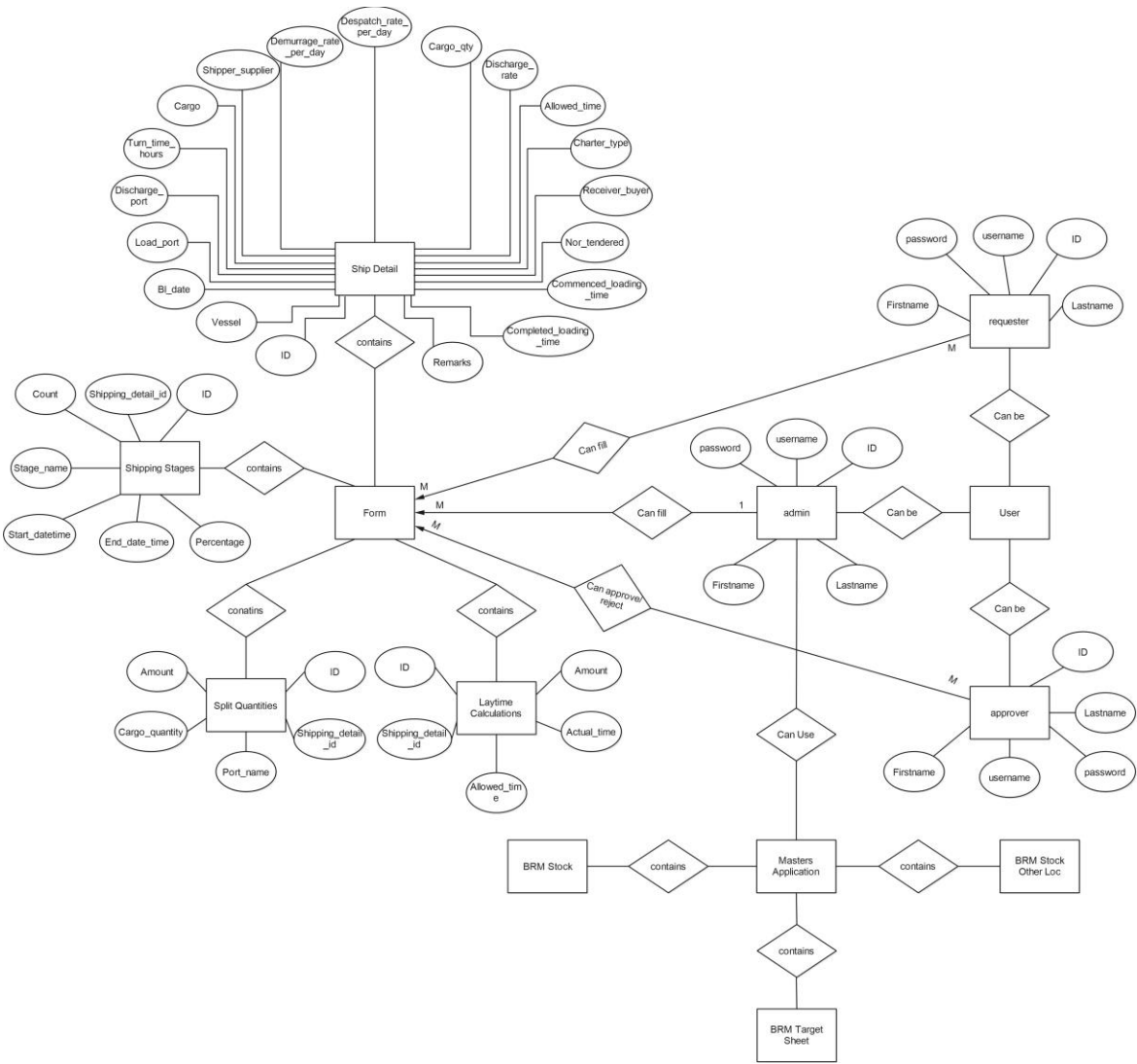


Figure 4.10 ER Diagram

CHAPTER 5 IMPLEMENTATION AND TESTING

5.1 Hardware and Software Requirements

Hardware Requirements:

- Computer with internet connectivity.
- Sufficient processing power and memory for smooth operation.
- Printer (optional) for printing reports.

Software Requirements:

- Operating System: Windows, Linux, or any other platform with a modern web browser.
- Web Browser: Chrome, Firefox, Edge, or any other modern web browser.
- Web Technologies: HTML, CSS, JavaScript.
- Web Server: Hosting environment supporting Django and Django REST framework.

5.2 Snapshots

➤ Login Page

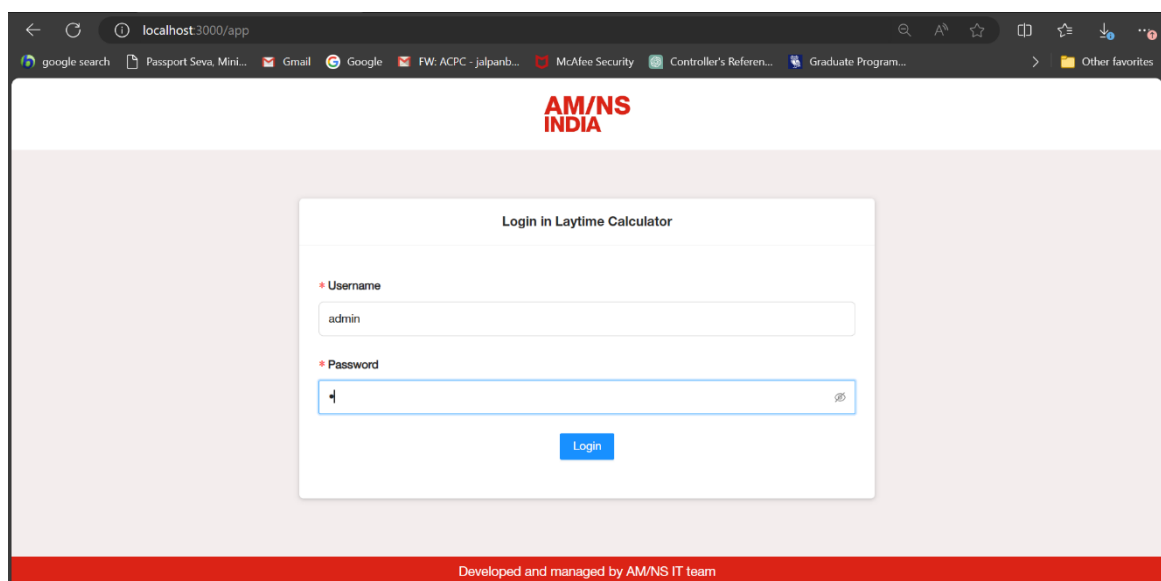


Figure 5.1 Login Page

- The figure represents the login interface where users enter their credentials to access the Laytime Calculator Website.

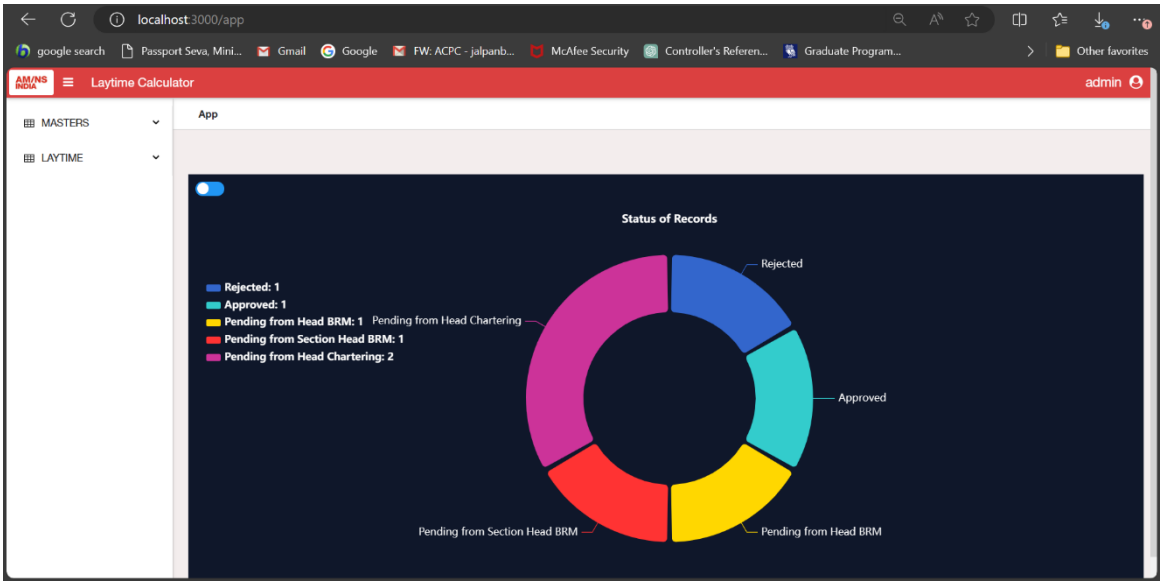


Figure 5.2 Dashboard

- The figure here displays the dashboard of the Laytime Calculator Website, where the user can see the Pending, Approved or Rejected records.

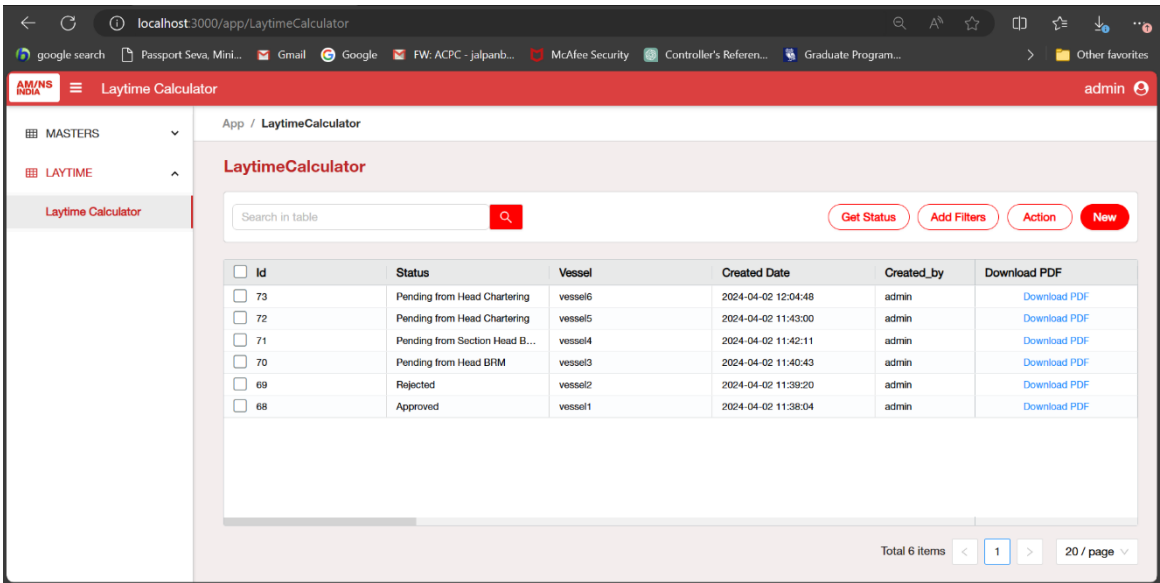


Figure 5.3 List View

- The Laytime Calculator subsection shows a list view interface in the Laytime Calculator Website, displaying data in a tabular format.

The screenshot shows a web application titled 'Laytime Calculator' with a sidebar menu containing 'MASTERS' and 'LAYTIME'. The 'LAYTIME' section is expanded, showing 'Laytime Calculator'. The main form area is titled 'Ship Detail' and contains the following fields:

Field	Value	Field	Value
Vessel	vessel7	* Demurrage rate/day (USD \$)	5000
B/L Date	03-04-2024	* Despatch rate/day (USD \$)	2500
Load Port	lport	* Cargo Qty (M/T)	2000
Discharge Port	dport	* Discharge rate	100

Buttons: Back, Actions, Clear

Figure 5.4 Despatch Case New Form (1)

The screenshot shows the same web application, but the form is now the 'Calculate' form. It contains the following fields:

Field	Value	Field	Value
Turn Time (hrs)	10	* Time allowed	20.00000
Cargo	cargo	Charter Type	charter
Shipper/Supplier	shipper	Receiver / Buyer	buyer
NOR Tendered	03-04-2024 17:02	Commenced Loading Time	04-04-2024 17:02
Completed Loading Time	18-04-2024 17:02	Remarks	Despatch

Buttons: Back, Actions, Clear, Calculate

Figure 5.5 Despatch Case New Form (2)

The screenshot shows the 'Shipping Stage details' form in the Laytime Calculator application. The form is titled 'Shipping Stage details' and has a 'Back' button and 'Actions' and 'Clear' buttons. It contains two rows of data for shipping stages. The first row is for 'Customs cleared' with a count of 1, start date of 04-04-2024 17:02, end date of 10-04-2024 17:02, and percentage of 100%. The second row is for 'Discharging' with a count of 1, start date of 05-04-2024 17:02, end date of 08-04-2024 17:02, and percentage of 100%. There is an '+ Add Shipping stage' button at the bottom. Below the shipping stage details is a section for 'Split Cargo Quantity' with fields for Port, Cargo Quantity, and Amount.

Count	Stage Name	Start Date Time	End Date Time	Percentage
1	Customs cleared	04-04-2024 17:02	10-04-2024 17:02	100%
1	Discharging	05-04-2024 17:02	08-04-2024 17:02	100%

Figure 5.6 Despatch Case New Form (3)

The screenshot shows the 'Split Cargo Quantity' form in the Laytime Calculator application. The form is titled 'Split Cargo Quantity' and has a 'Back' button and 'Actions' and 'Clear' buttons. It contains two rows of data for cargo quantity splits. The first row is for 'Hazira' with a cargo quantity of 123 and an amount of 1691.25. The second row is for 'Vizag' with a cargo quantity of 333 and an amount of 4578.75. There is an '+ Add Quantity Split' button at the bottom. Below the cargo quantity splits is a section for 'Remaining Cargo Quantity' with a value of 1544.

Port	Cargo Quantity	Amount
Hazira	123	1691.25
Vizag	333	4578.75

Remaining Cargo Quantity: 1544

Figure 5.7 Despatch Case New Form (4)

- Here **Figures from 5.4 to 5.7** represent the initial step or interface for creating a new form in the Laytime Calculator Website. The figures are an example of the **Despatch case**.

AM/NS INDIA | Laytime Calculator | admin

MASTERS | LAYTIME | Laytime Calculator

Back | Actions | Clear

Completed Loading Time: 18-04-2024 17:02

Remarks: Despatch

Calculate

Laytime Calculation

Total Allowed time (in days)	Total Actual time (in days)	Time Saved (in days)	Total Despatch Amount
20	9	11	27500

Figure 5.8 Calculation (Despatch)

- The table in the form displays the calculation process for **Despatch** in the Laytime Calculator Website.

AM/NS INDIA | Laytime Calculator | admin

MASTERS | LAYTIME | Laytime Calculator

Back | Actions | Clear

Completed Loading Time: 18-04-2024 17:02

Remarks: Despatch

Calculate

Laytime Calculation

Total Allowed time (in days)	Total Actual time (in days)	Time Saved (in days)	Total Despatch Amount
20	9	11	27500

Figure 5.9 Despatch Case Save Form

- The figure shows the process of saving a form in the Laytime Calculator Website by hovering on the Actions button. Initially only Save button will appear.

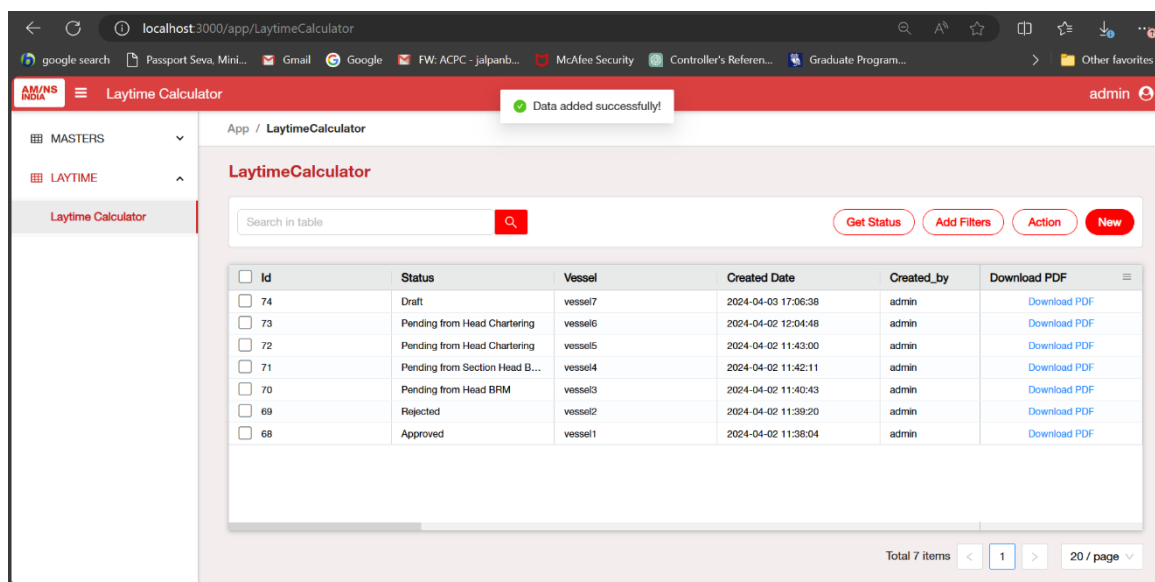


Figure 5.10 Despatch Case Data Saved Successfully

- The user will be redirected to the List View and a success message will appear after saving data in the Laytime Calculator Website.

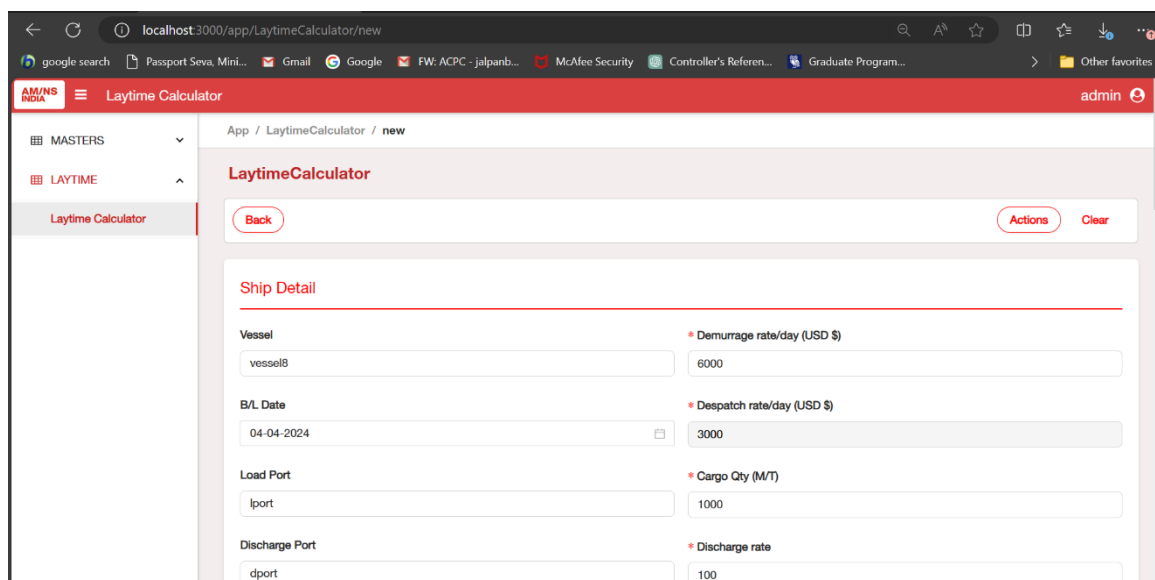


Figure 5.11 Demurrage Case New Form (1)

The screenshot shows a web application titled "Laytime Calculator" with a sidebar menu containing "MASTERS" and "LAYTIME". The "LAYTIME" section is expanded, showing "Laytime Calculator". The main form has a "Back" button and "Actions" and "Clear" buttons. The form fields are as follows:

General Details	
Turn Time (hrs)	* Time allowed
20	10.00000
Cargo	Charter Type
cargo	charter
Shipper/Supplier	Receiver / Buyer
shipper	buyer
NOR Tendered	Commenced Loading Time
04-04-2024 17:08	05-04-2024 17:08
Completed Loading Time	Remarks
20-04-2024 17:08	Demurrage

A "Calculate" button is located at the bottom of the form.

Figure 5.12 Demurrage Case New Form (2)

The screenshot shows the "Shipping Stage details" section of the "Laytime Calculator" form. It contains a table with the following data:

Count	* Stage Name	* Start Date Time	* End Date Time	* Percentage
☒	Discharging	05-04-2024 17:09	20-04-2024 17:09	100%
☒	Discharging commenced	05-04-2024 20:09	08-04-2024 17:09	100%

Below the table is a "+ Add Shipping stage" button. The "Split Cargo Quantity" section is also visible, with fields for "Port", "Cargo Quantity", and "Amount".

Figure 5.13 Demurrage Case New Form (3)

The screenshot shows a web application titled 'Laytime Calculator' with a sidebar menu containing 'MASTERS' and 'LAYTIME'. The 'LAYTIME' section is expanded, showing 'Laytime Calculator'. The main form is titled 'Split Cargo Quantity' and includes a 'Back' button and 'Actions' and 'Clear' buttons. It features two rows of input fields for 'Port', 'Cargo Quantity', and 'Amount'. The first row has 'Vizag' for Port, '234' for Cargo Quantity, and '11056.5' for Amount. The second row has 'Paradip' for Port, '445' for Cargo Quantity, and '21026.25' for Amount. Below these is a '+ Add Quantity Split' button and a 'Remaining Cargo Quantity' field showing '321'.

* Port	* Cargo Quantity	* Amount
Vizag	234	11056.5
Paradip	445	21026.25

+ Add Quantity Split

Remaining Cargo Quantity: 321

Figure 5.14 Demurrage Case New Form (4)

- Here **Figures from 5.11 to 5.14** represent the initial step or interface for creating a new form in the Laytime Calculator Website. The figures are an example of the **Desmurrage** case.

The screenshot shows the 'Calculation (Demurrage)' form. It includes a 'Back' button and 'Actions' and 'Clear' buttons. The form has two rows of input fields for 'Completed Loading Time' and 'Remarks'. The first row has '04-04-2024 17:08' for Completed Loading Time and '05-04-2024 17:08' for Remarks. The second row has '20-04-2024 17:08' for Completed Loading Time and 'Demurrage' for Remarks. Below these is a 'Calculate' button. The 'Laytime Calculation' section displays a table with four columns: 'Total Allowed time (in days)', 'Total Actual time (in days)', 'Time Exceed (in days)', and 'Total Demurrage Amount'. The values are 10, 17.875, 7.875, and 47250 respectively.

Total Allowed time (in days)	Total Actual time (in days)	Time Exceed (in days)	Total Demurrage Amount
10	17.875	7.875	47250

Figure 5.15 Calculation (Demurrage)

- The table in the form displays the calculation process for **Demurrage** in the Laytime Calculator Website.

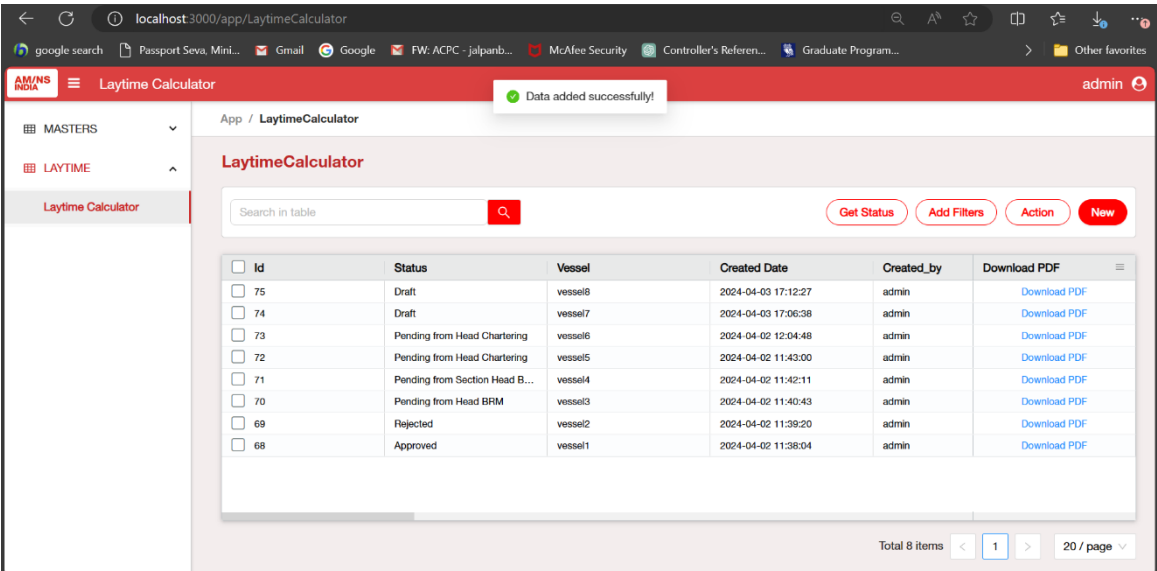


Figure 5.16 Demurrage Case Data added successfully

- The figure indicates a success message or confirmation after adding demurrage case data in the Laytime Calculator Website.

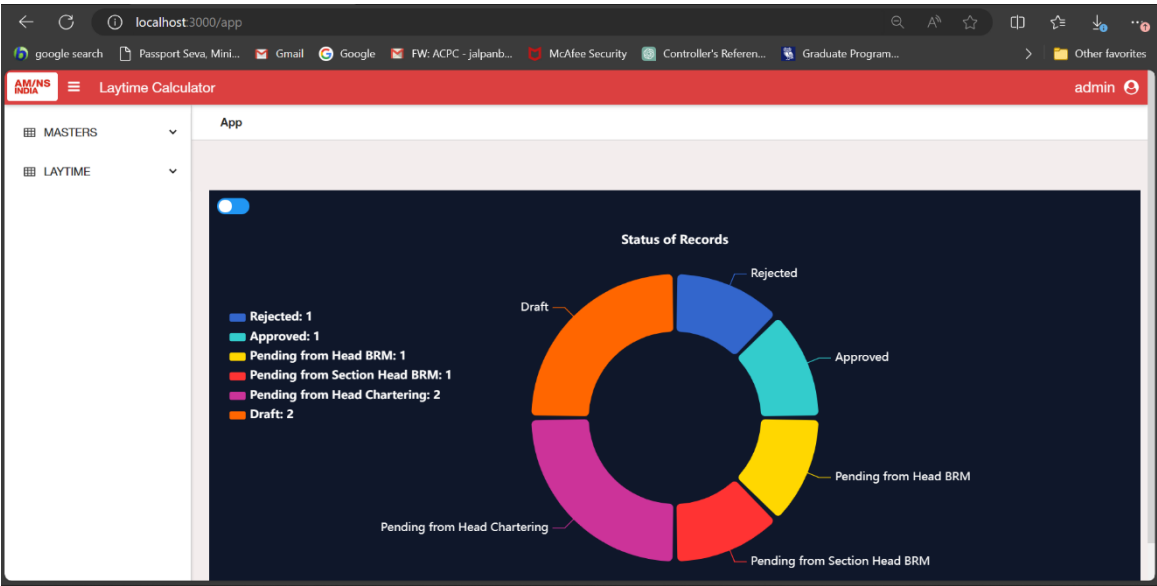


Figure 5.17 Updated Dashboard

- This figure shows the updated dashboard of the Laytime Calculator Website. Here, Both the new records are added as Draft in the Dashboard.

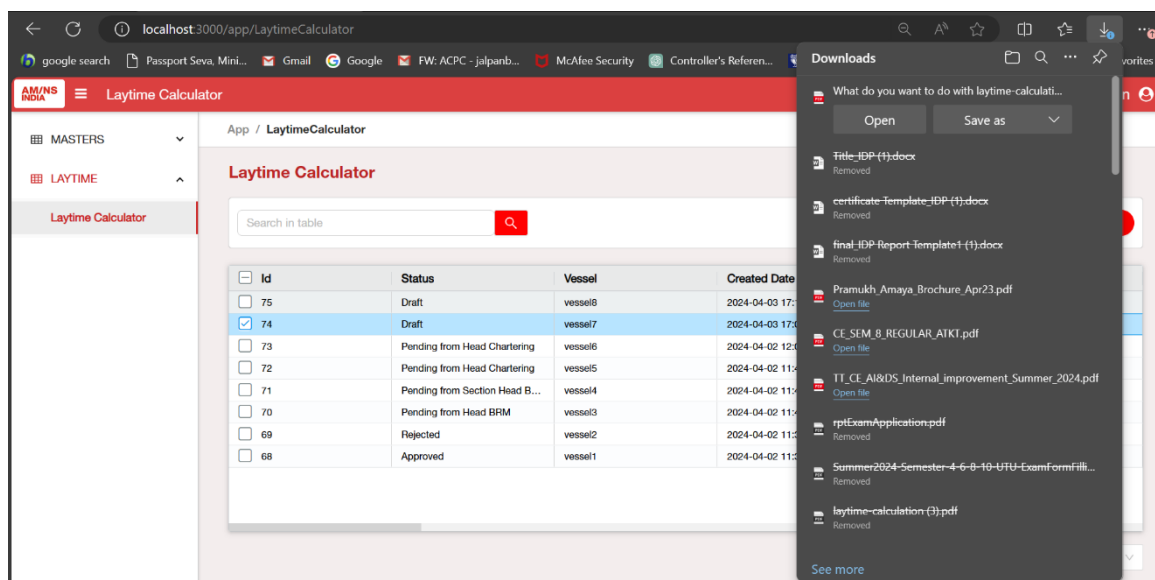


Figure 5.18 Download PDF

- Here after clicking on the 'Download PDF' button of a specific record, a pop-up will appear which will allow the user to either save or open the PDF.

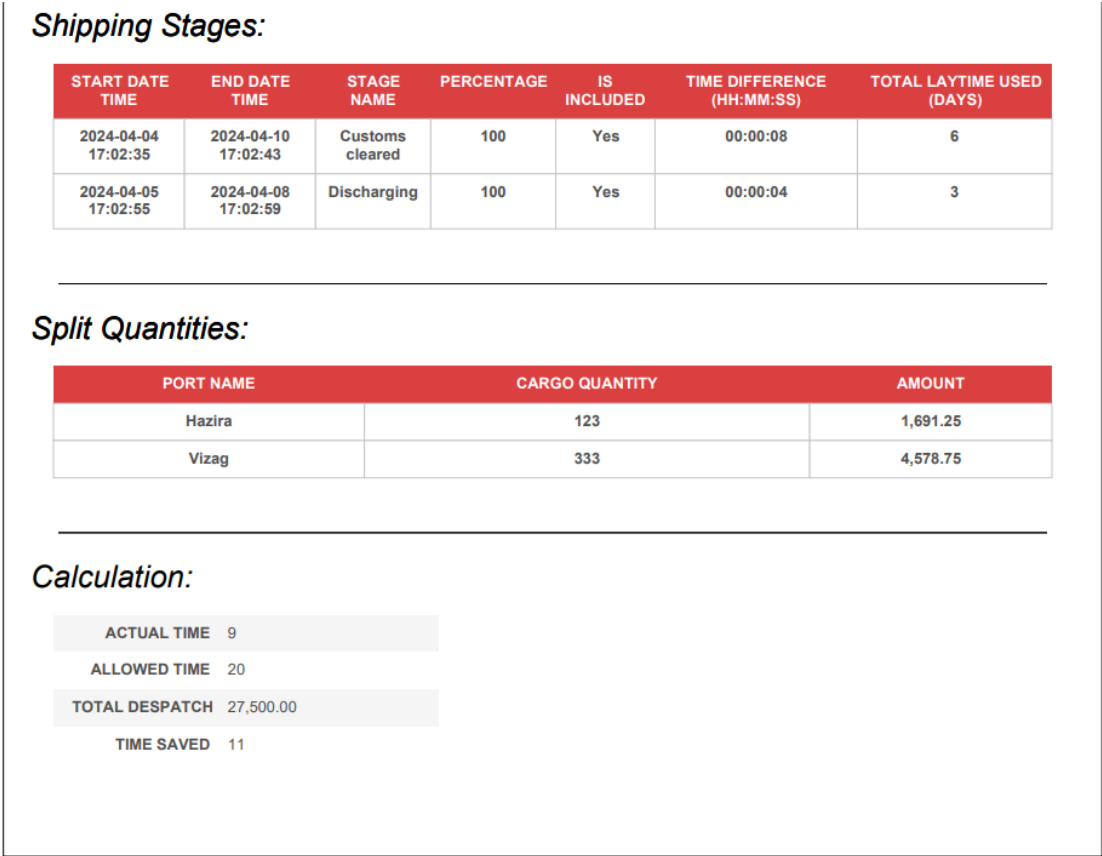
**AM/NS
INDIA**

Ship Detail:

NOR TENDERED	2024-04-03 17:02:08	SHIPPER	shipper
COMMENCED LOADING TIME	2024-04-04 17:02:12	SUPPLIER	
COMPLETED LOADING TIME	2024-04-18 17:02:19	DEMURRAGE RATE PER DAY	5000.00
VESSEL	vessel7	DESPATCH RATE PER DAY	2500.00
BL DATE	2024-04-03	CARGO QTY	2000
LOAD PORT	lport	DISCHARGE RATE	100
DISCHARGE PORT	dport	ALLOWED TIME	20
TURN TIME HOURS	10	CHARTER TYPE	charter
CARGO	cargo	RECEIVER BUYER	buyer
		REMARKS	Despatch

Figure 5.19 PDF (1)

- The PDF will contain Ship Details, Shipping Stages, Quantity Split, and Laytime Calculator. This figure shows the first page of the PDF generated from the Laytime Calculator Website.



Page 1 of 1

Figure 5.20 PDF (2)

This figure shows the second page of the PDF generated from the Laytime Calculator Website.

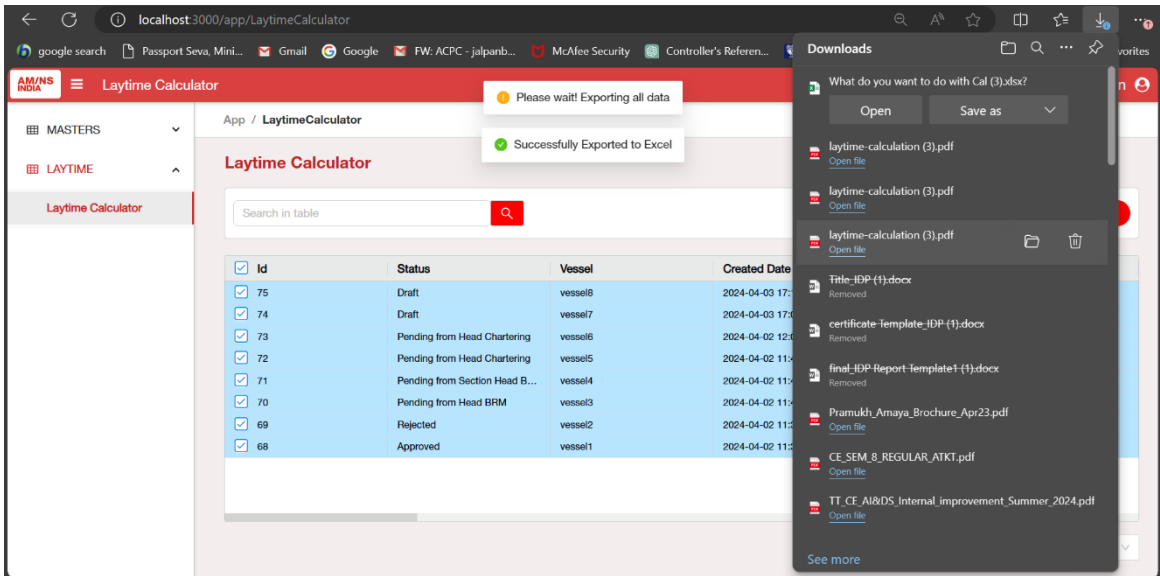


Figure 5.21 Export Excel

- Here, the figure displays the option to export the selected data from the Laytime Calculator Website to an Excel file.

ID	STATUS	CREATED_BY	CREATED_DATE	VESSEL	BL_DATE	LOAD_PORT	DISCHARGE_PORT	TURN_TIME_HOURS	CARGO	SHIPPER_SUPPLIER	DEMURRAGE_RATE_PER_DAY
73	Pending from Head Chartering	admin	2024-04-02T12:04:48.954151	vessel6	2024-04-03	lport	dport	10	cargo	shipper	16000.00
74	Draft	admin	2024-04-03T17:06:38.865873	vessel7	2024-04-03	lport	dport	10	cargo	shipper	5000.00
75	Draft	admin	2024-04-03T17:12:27.126838	vessel8	2024-04-04	lport	dport	20	cargo	shipper	6000.00
68	Approved	admin	2024-04-02T11:38:04.428740	vessel1	2024-04-02	lport	dport	10	cargo	shipper	12000.00
69	Rejected	admin	2024-04-02T11:39:20.535502	vessel2	2024-04-03	lport	dport	10	cargo	shipper	10000.00
71	Pending from Section Head BRM	admin	2024-04-02T11:42:11.968427	vessel4	2024-04-03	lport	dport	10	cargo	shipper	14000.00
72	Pending from Head Chartering	admin	2024-04-02T11:43:00.047041	vessel5	2024-04-02	lport	dport	10	cargo	shipper	15000.00
70	Pending from Head BRM	admin	2024-04-02T11:40:43.962226	vessel3	2024-04-02	lport	dport	10	cargo	shipper	11000.00

Figure 5.22 Excel

- This image shows the exported Excel file containing data from the Laytime Calculator Website.

Id	Status	Vessel	Created Date
73	Pending from Head Chartering	vessel6	2024-04-02 12:04:48
72	Pending from Head Chartering	vessel5	2024-04-02 11:43:00

Figure 5.23 Filter Status

- After hovering on the 'Get Status' button, all the status of records will appear and the user can select any status and those records will be displayed.

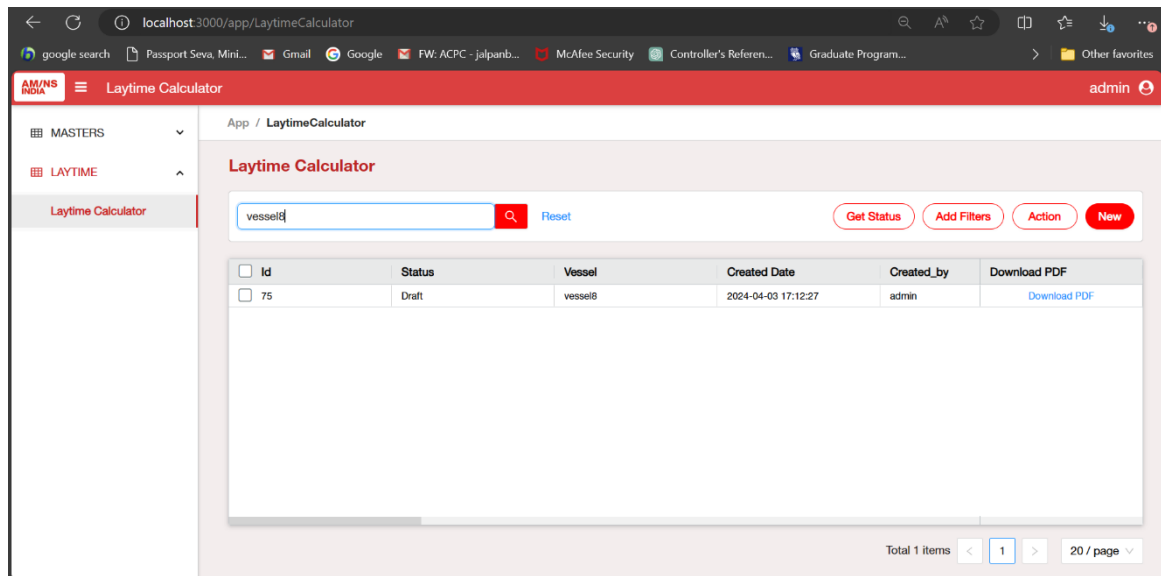


Figure 5.24 Search

- Here the user can use the generic search feature which will allow the user to get the record containing the word that they are searching for.

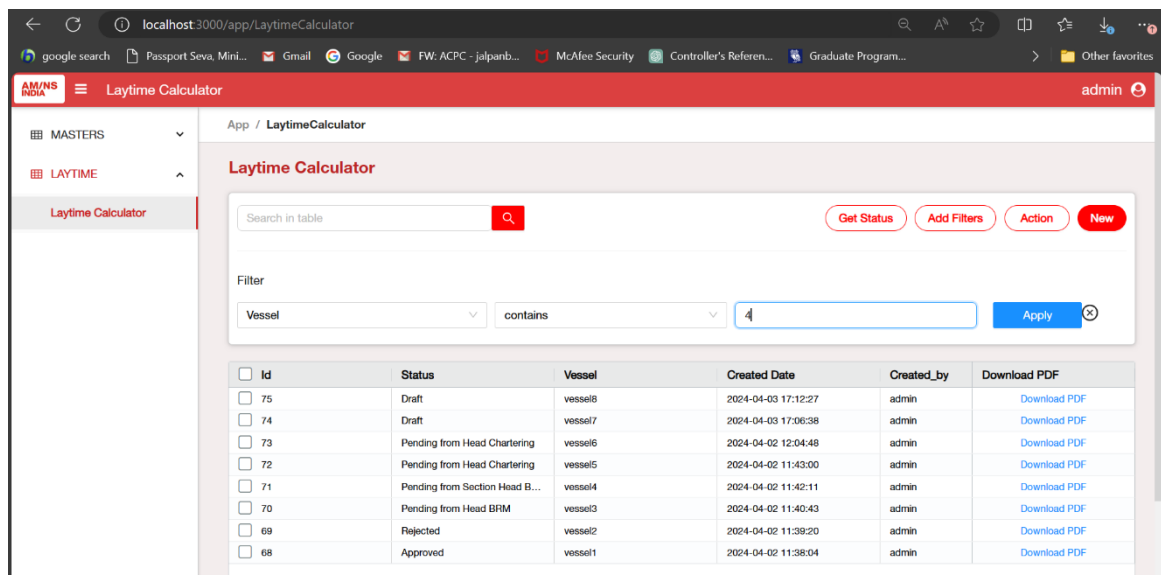


Figure 5.25 Apply filter

- A filter can also be used that can be applied to any specific column.

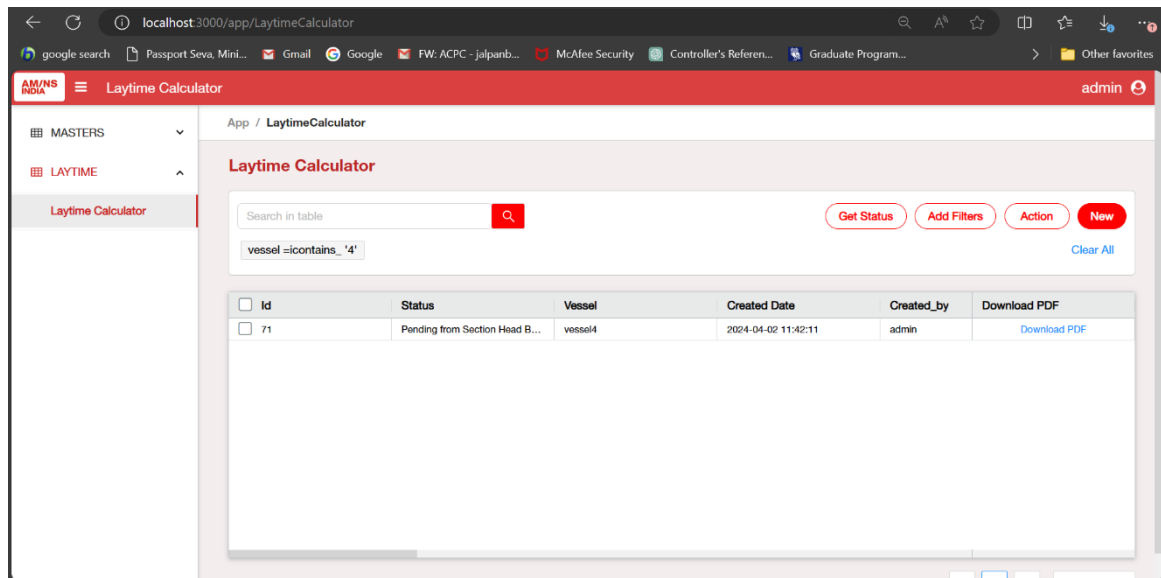


Figure 5.26 Filter Result

- This image shows the filtered record in the list view.

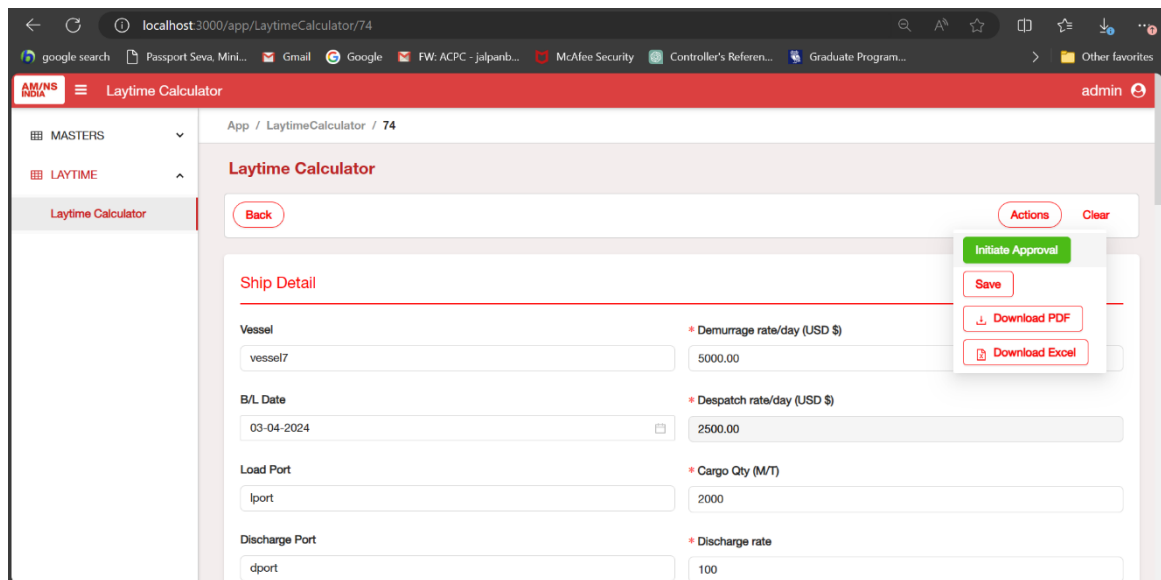


Figure 5.27 Initiate Approval Workflow

- Only after saving a record, the Actions dropdown will have the Initiate Approval button and Download button. Here the user can click on the Initiate Approval button to allow the approver to Approve or Reject that specific record.

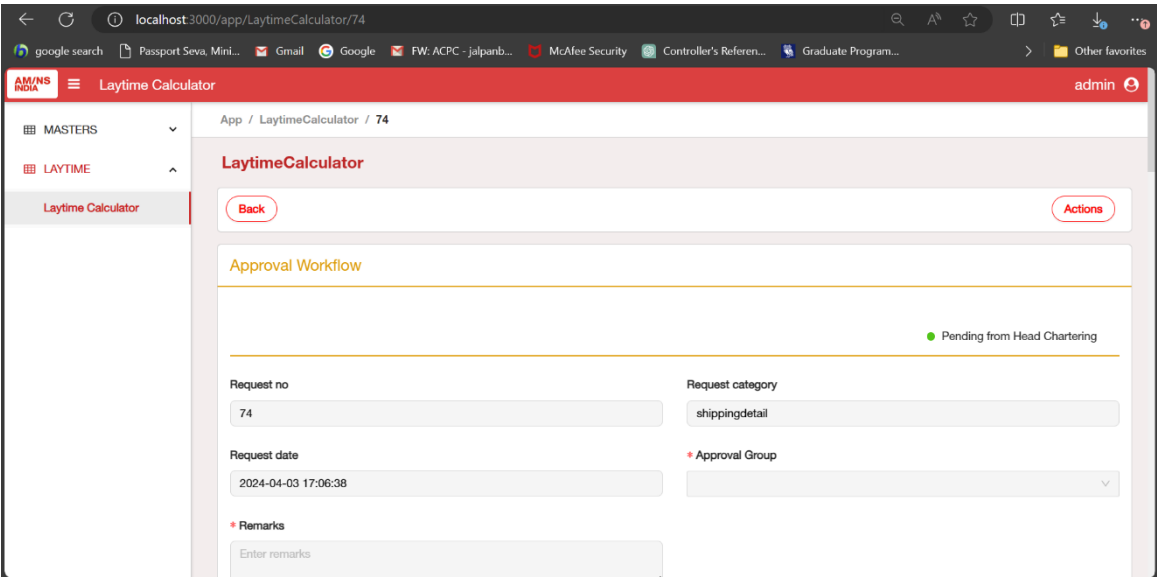


Figure 5.28 Approval Form Initiated

- The Figure shows the Approval Form and it contains Request id, Remarks section, Approval Group which will have who has Approved/ Rejected it, and a Request date which indicates on which date and time the record was created.

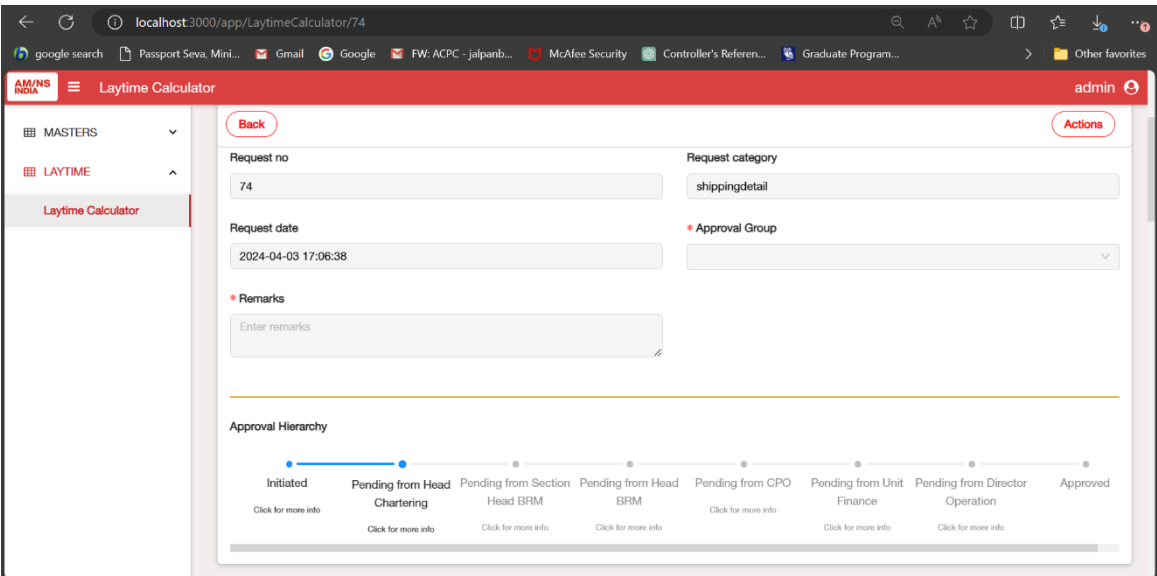


Figure 5.29 Approval Hierarchy

- The requester and approver both can see the hierarchy and see who has approved or rejected.

The screenshot shows the 'Laytime Calculator' web application. The left sidebar has a menu with 'MASTERS' and 'LAYTIME'. The 'LAYTIME' section is expanded, showing 'Laytime Calculator'. The main content area has a 'Back' button and an 'Actions' button. The form includes fields for 'Request no' (77), 'Request category' (shippingdetail), 'Request date' (2024-04-12 11:24:19), and 'Remarks'. Below these is an 'Approval Hierarchy' section showing a flowchart with steps: Initiated, Pending from Head Chartering, Pending from Section Head BRM, Pending from Head BRM, Pending from CPO, Pending from Unit Finance, Pending from Director Operation, Pending from CEO, and Approved. The 'Ship Detail' section is partially visible at the bottom.

Figure 5.30 Approval Hierarchy with the CEO as the last approver

- Here the hierarchy is different and has the CEO as the last approver because for the amount is between 1 crore and 5 crore.

The screenshot shows an email template titled 'LAYTIME DETAILS' from AM/NS INDIA. The email body contains a greeting 'Dear Sir/Madam,' and a request to find details for request #75. It includes a table for 'Request ID', 'Current Status', 'Initiator', and 'Remark'. Below this is a section for 'The laytime calculations for request #75 are as follows:' which contains three tables: 'Ship Details', 'Stages', and 'Laytime Calculations'.

Request ID	Current Status	Initiator	Remark
#75	Pending from Head Chartering	admin	Initiated

The laytime calculations for request #75 are as follows:

Ship Details																	
NOR Tendered	Commenced Loading Time	Completed Loading Time	Vessel	B/L Date	Load Port	Discharge Port	Turn Time (hrs)	Cargo	Shipper/Supplier	Demurrage rate/day (USD \$)	Despatch rate/day (USD \$)	Cargo Qty (MT)	Discharge rate	Time allowed	Charter Type	Receiver/Buyer	Remarks
2024-04-04 17:08:34	2024-04-05 17:08:46	2024-04-20 17:08:53	vessel8	2024-04-04	lport	dport	20.0	cargo	shipper	6000.00	3000.00	1000.0	100.0	10.0	charter	buyer	Demurrage

Stages				Quantity Split			Laytime Calculations			
Stage Name	Start Date Time	End Date Time	Percentage	Port Name	Cargo Quantity	Amount	Actual time (days):	Allowed time (days):	Total demurrage:	Time exceeded (days):
Discharging	2024-04-05 17:09:00	2024-04-20 17:09:10	100	Vizag	234.0	11,056.50	17.875	10.0	47,250.00	7.875
Discharging commenced	2024-04-05 20:09:19	2024-04-08 17:09:25	100	Paradip	445.0	21,026.25				

Buttons: Approve/Reject

For additional information or actions, kindly get in touch with your specific team.

Regards,

Figure 5.31 Email when the approval form is initiated

- The figure shows the email template when an approval form is initiated in the Laytime Calculator Website.

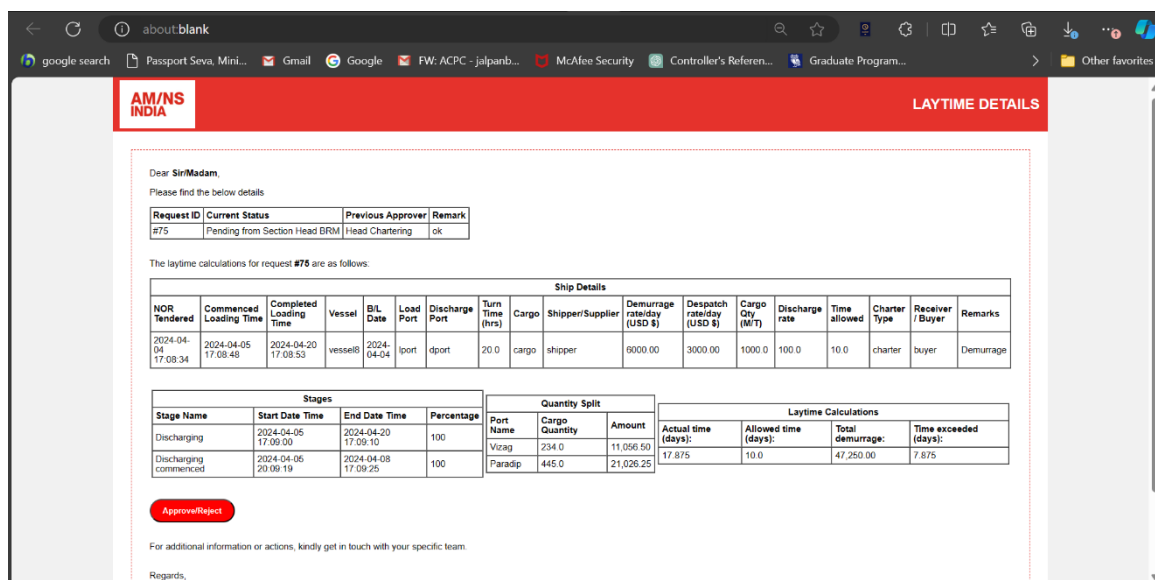


Figure 5.32 Email when approved

- The figure shows the email template when a record is approved by an approver in the Laytime Calculator Website.

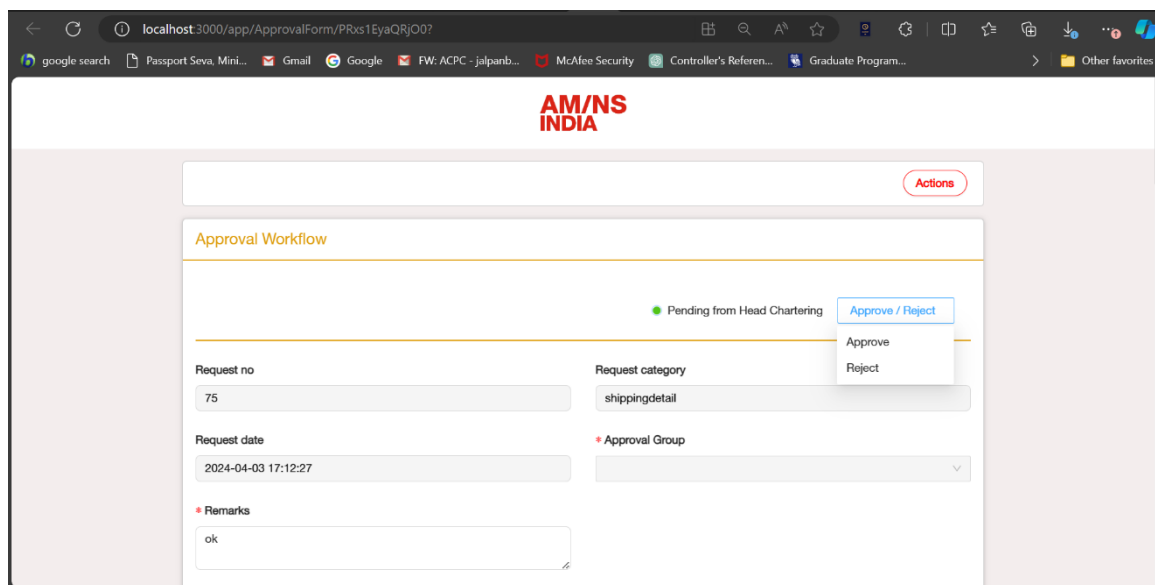


Figure 5.33 Approval Form

- This approval form will appear after clicking on the 'Approve/Reject' button in the email. The approver can approve or reject a record here **without logging in**.

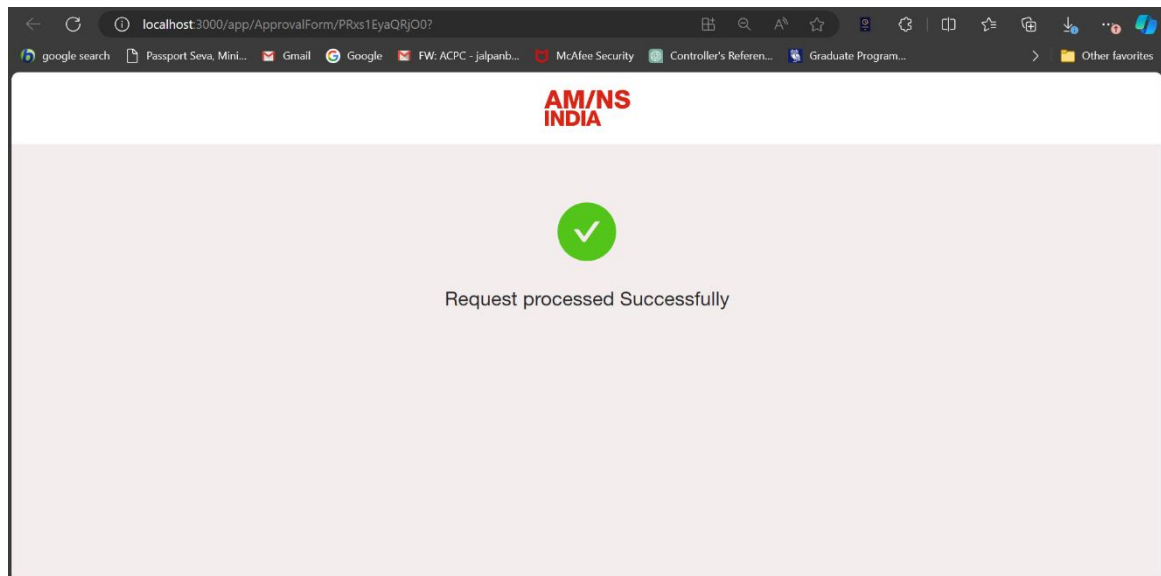


Figure 5.34 The request is processed successfully

- If any action is taken on the record that is if the approver either Approves or Rejects a record then a message will appear stating that the request has been processed successfully.

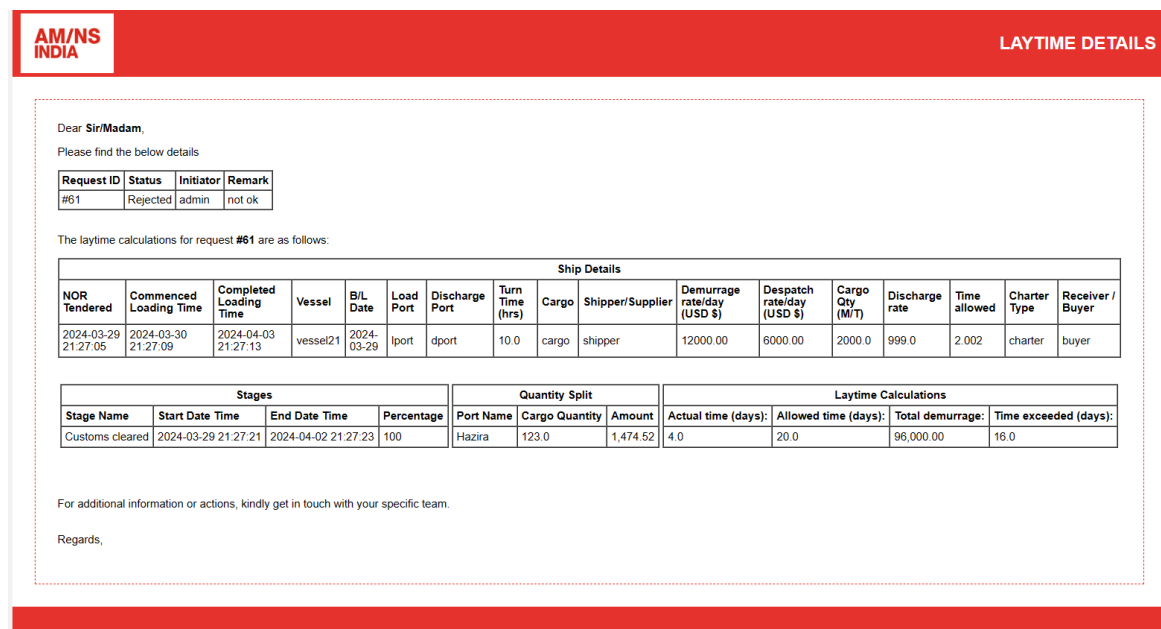


Figure 5.35 Email for Rejected case

- The figure shows the email template for a case when the approver rejects a record. Here an email will be sent to the initiator and the 'Approver/Reject' button will disappear

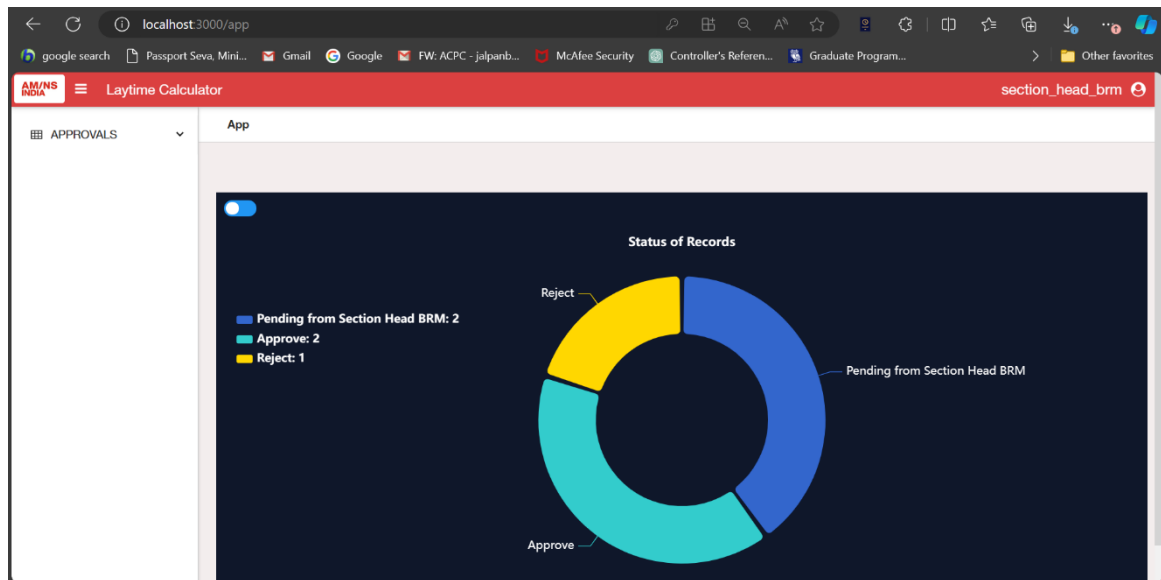


Figure 5.36 Approver's Dashboard

- This figure shows the Dashboard of an Approver after the approver logs in.

The table displays the following records:

Id	Status	Vessel	Created Date	Created_by	Download PDF
75	Pending from Section Head B...	vessel6	2024-04-03 17:12:27	admin	Download PDF
71	Pending from Section Head B...	vessel4	2024-04-02 11:42:11	admin	Download PDF

Total 2 items | 1 / 20 / page

Figure 5.37 Approver List View

- In the approver's list view, there will be records with status Pending from their end.

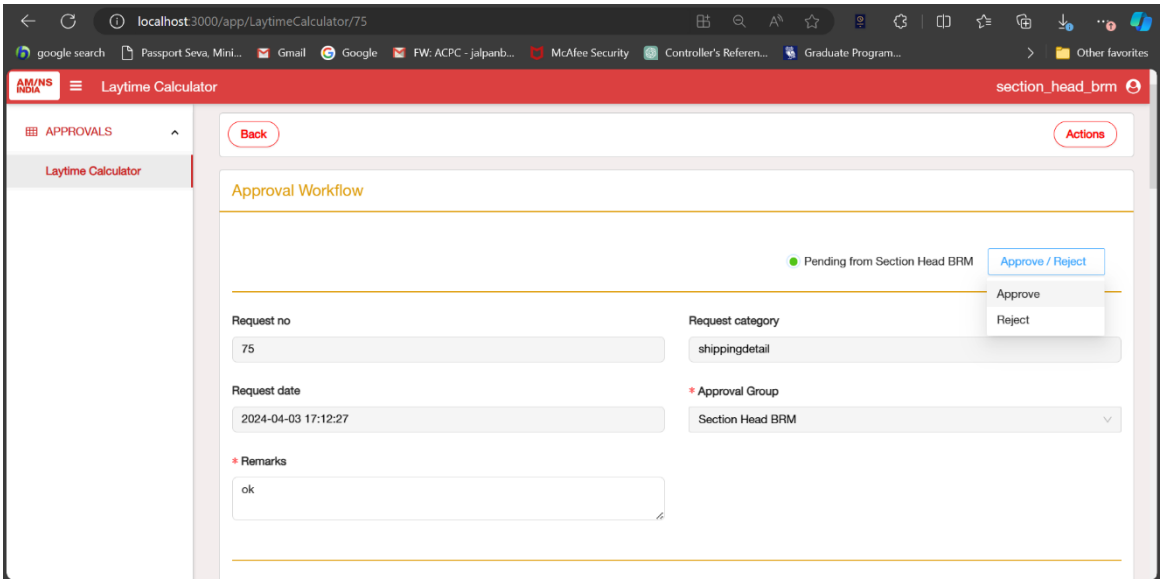


Figure 5.38 Approving/Rejecting after logging in

- Here the approver can also see the Approval form by clicking on the record and do the Approval process.

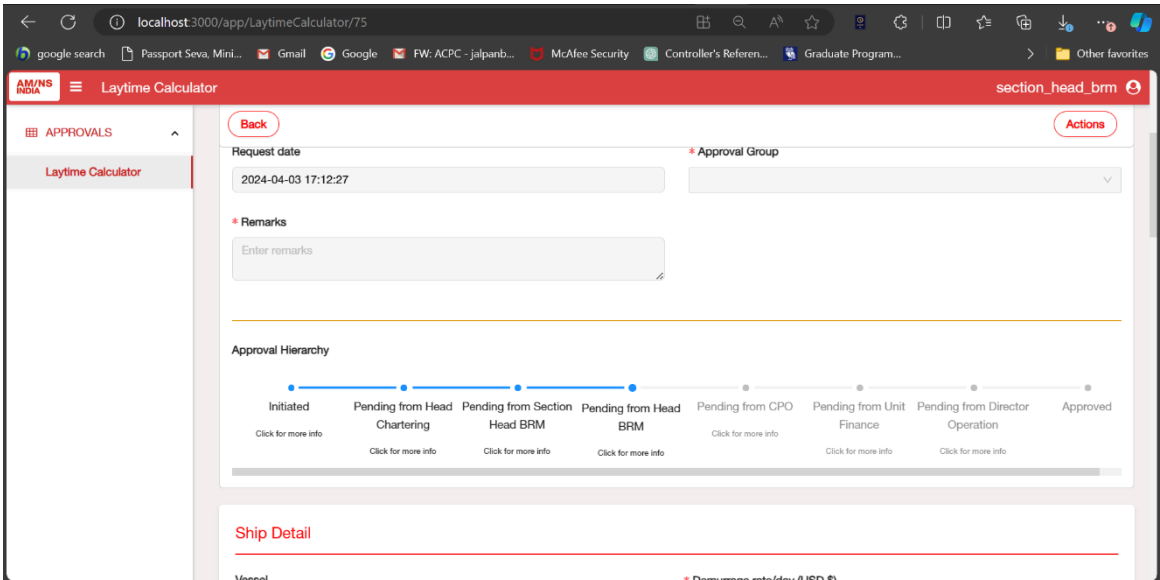


Figure 5.39 Updated Hierarchy

- The approver can also view the hierarchy flow.

App / Mstbrmstock

BRM Stock

Search in table

Add Filters Action Now

☐	Id	Material ref	Raw report	Material	Rwb	Total stock name
☐	36	INGWE	INGWE S.A.	Corex Coal	Blacks	IGWA COAL
☐	35	Thermal Coal	Thermal Coal Indonesia	Power Coal	Blacks	THERMAL COAL
☐	34	Poland BF Coke	BF Poland	Met Coke	Blacks	POLAND COKE
☐	33	Russian BF Coke	BF Russia	Met Coke	Blacks	RUSSIAN COKE
☐	32	BPCI	BPCI Russia	Corex Coal	Blacks	RUSSIAN PCI CO/
☐	31	VSKPCI	VSKPCI Russia	Corex Coal	Blacks	RUSSIAN PCI CO/
☐	30	Steam Coal	Steam Coal	Corex Coal	Blacks	RUSSIAN STEAM
☐	29	Japan BF Coke	BF Japan	Met Coke	Blacks	JAPAN COKE
☐	28	Poland nut Coke	CX Poland	Corex Coke	Blacks	POLLAND NUT CC
☐	27	BF_Pdp	BF Paradeep	Pellets	Reds	BF PARADEEP PLI
☐	26	BF_Pdp	BF Paradeep	Pellets	Reds	BF PARADEEP PLI

Total 36 items < 1 2 > 20 / page Go to Page

Figure 5.40 Masters Application BRM Stock List View

- The Admin can also access the MASTERS application which will contain three subsections BRM stocks, BRM Stock Other Loc, and BRM Target Sheet. This figure shows BRM Stock's List View.

App / Mstbrmstock / new

BRM Stock

Back Clear Save

BRM Stock Master

Material Ref: materialref3 Raw Report: report3

material: material3 RWB: rwb3

Total Stock Name: stock3 Ground Stock: 1234

Figure 5.41 Masters Application BRM Stock Form

- The Figure here shows BRM Stock's form fields.

App / Mstbrmstockotherloc

BRM Stock Other Loc

Search in table

Add Filters Action Now

☐	Id	Material ref	Unit	Dabuna	Paradeep	Kirandul
☐	9	materialref1	123	10	20	30
☐	8	materialref8	345	1234	455	345
☐	7	BF_Pdp	DMT	0	173213	0
☐	6	DR_Pdp	DMT	0	0	0
☐	5	DR_Vzg	WMT	0	0	0
☐	4	Oxide Fines	WMT	770169	40339	0
☐	3	Mill Scale	WMT	0	11116	0
☐	2	Concentrate Slurry	DMT	12560	0	11248
☐	1	Thermal Coal	MT	0	22409	0

Total 9 items < 1 > 20 / page

Figure 5.42 Masters Application BRM Stock Other Loc List View

➤ Here the figure displays BRM Stock Other Loc's list view.

App / Mstbrmstockotherloc / new

BRM Stock Other Loc

Back Clear Save

BRM Stock Other Loc Master

Material Ref	Unit
materialref8	345
Dabuna	Paradeep
1234	455
Kirandul	Vizag
345	124

Figure 5.43 Masters Application BRM Stock Other Loc Form

➤ The Figure here shows BRM Stock Other Loc's form fields.

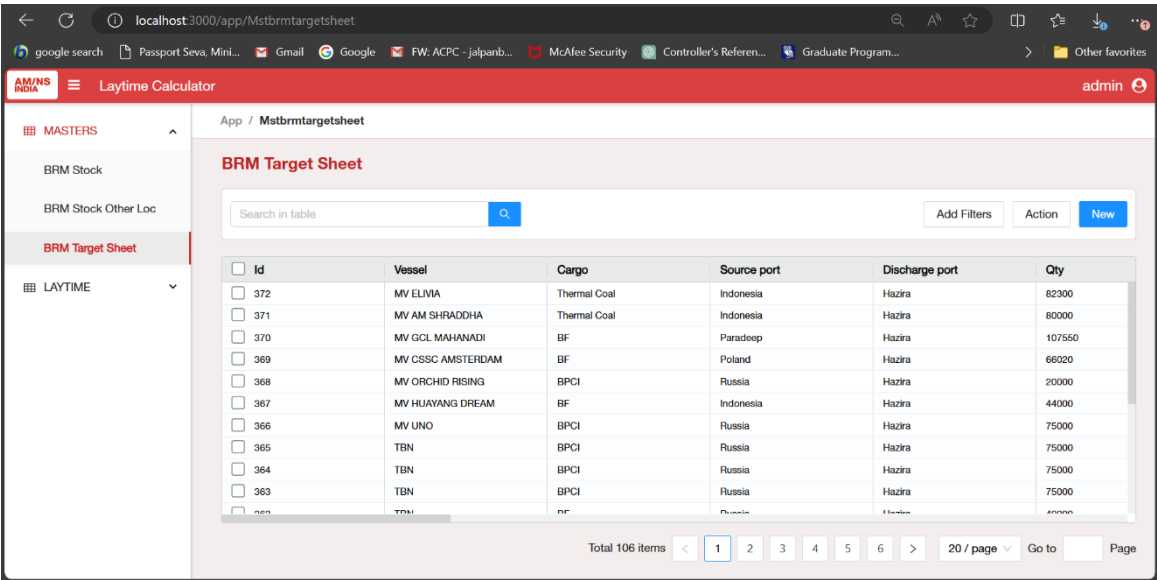


Figure 5.44 Masters Application BRM Target Sheet List View

➤ The figure displays BRM Target Sheet List View.

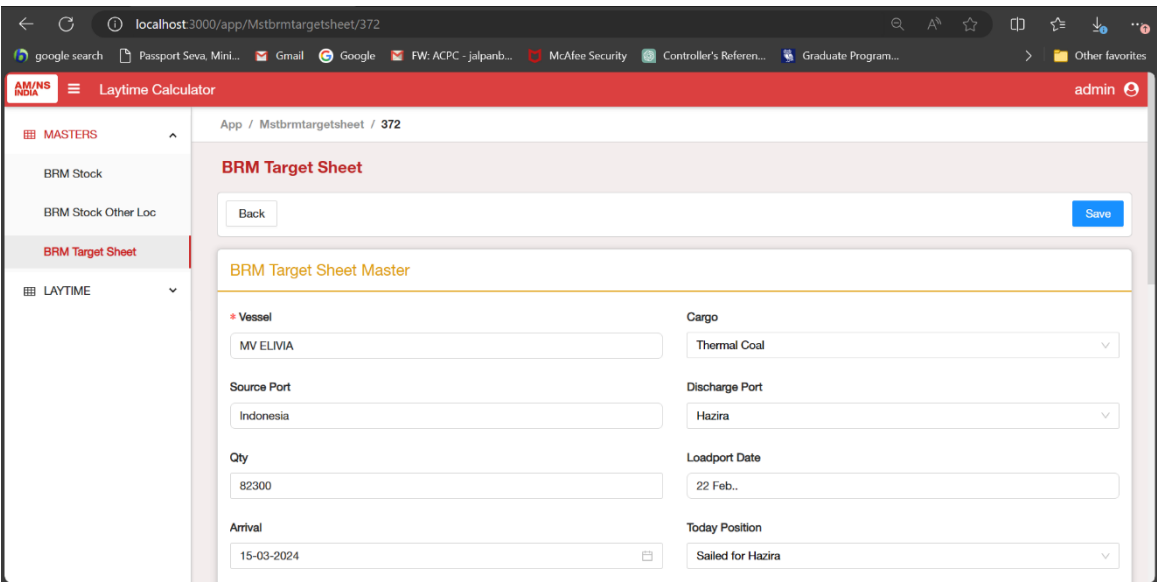


Figure 5.45 Masters Application BRM Target Sheet Form

➤ This figure shows BRM Target Sheet form fields.

5.3 Test Cases

Table 5.1 Test Cases

TestID	Case	Test Data	Expected Result	Actual Result	Pass/Fail
1	Login	- Internet Connectivity - Login ID - Password	If the Login ID & password are empty or invalid and No Internet connectivity then display an Error message otherwise login successfully	If the Login ID & password are empty or invalid and no Internet connectivity then display an error message.	Pass
2	Dashboard View	- Data availability	If data is available then display the dashboard with Pending, Approved, and Rejected records for the specific user else display the message 'No Data Available'.	If data is available then the dashboard with Pending, Approved, and Rejected records for the specific user is displayed else a message 'No Data Available' appears.	Pass
3	Adding New Form	- Ship Details - Shipping Stages - Quantity Split	Successfully add a new form with all necessary data included if correct data is added else print message to correct field values whenever needed	Successfully add a new form with all necessary data included if correct data is added else print message to correct field	Pass

				values whenever needed	
4	Calculate Despatch or Demurrage	- Time Allowed to discharge - Shipping stage start datetime - Shipping stage end datetime	A calculation table will appear with columns -Total Allowed time (in days). -Total Actual time (in day). -Time Saved (in days) [For Despatch case] -Time Exceeded (in days) [For Demurrage case] -Total Despatch Amount [For Despatch case] -Total Demurrage Amount [For Demurrage case]	A calculation table will appear with columns -Total Allowed time (in days). -Total Actual time (in day). -Time Saved (in days) [For Despatch case] -Time Exceeded (in days) [For Demurrage case] -Total Despatch Amount [For Despatch case] -Total Demurrage Amount [For Demurrage case]	Pass
5	Saving Form	- Form data	If data is correct in the form then Save the form data and redirect to the List View with a	If data is correct in the form then Save the form data and redirect to	Pass

			success message else display error or warning message.	the List View with a success message else display error or warning message.	
6	Generate PDF	- Record id - Ship Details - Shipping Stages - Quantity Split - Laytime Calculations	Download the PDF file containing detailed information about the selected record	PDF file containing record details generated	Pass
7	Exporting Data to Excel	- Selected Record id - Column name	Export the selected data from the List View to an Excel file	Data exported successfully to an Excel file	Pass
8	Update Record	- Updated record data	If the updating is done successfully and changes are saved then display the message 'Data updated successfully' else give the error message 'Failed to update Data'	If the updating is done successfully and changes are saved then display the message 'Data updated successfully' else give the error message 'Failed to update Data'	Pass
9	Initiating Approval	- Saved record data	Display the Approval Form	Displayed the Approval	Pass

	Workflow		after initiating the approval workflow	Form after initiating the approval workflow	
10	Approving Record	- Remarks - Request id - Approval Group	Approve or Reject the record and update the status accordingly	Record approved or rejected and status updated accordingly	Pass
11	Email Notification (Approval Process)	- Approval status	Send email notifications to users based on the approval status	Email notifications are sent based on the approval status	Pass
12	Filtering Records	- Filter criteria	Apply the filter criteria to display specific records in the List View	Filter applied successfully and specific records displayed in the List View	Pass
13	Search records	- Search keyword	Search for a keyword and display records containing the keyword in any column	Keywords are searched successfully and relevant records are displayed in the List View	Pass
14	Logout	Select logout	The user should logout from the application and be redirected to the login page.	The user logs out from the application and is redirected to the login page.	Pass

CHAPTER 6 CONCLUSION AND FUTURE SCOPE

The Laytime Calculator Application is designed to streamline laytime calculations for vessel operations with efficiency and precision. It offers users a user-friendly interface, enabling seamless input of crucial information such as vessel details, cargo specifics, and important dates. With dynamic stage addition capabilities, users can accurately represent various segments of vessel operations. Real-time calculations provide instant insights into stage durations and total laytime, allowing for quick identification of extra time or demurrage. The application integrates seamlessly with a relational database, ensuring secure storage of calculation data and facilitating easy export and generation of Demurrage reports for further analysis. Moreover, the application supports business intelligence endeavors by leveraging historical laytime records for trend analysis and informed decision-making. Additionally, features such as support for amendment cases and customizable percentages enhance the application's flexibility and adaptability to diverse scenarios. Connectivity with other databases enhances data integration capabilities, while the implementation of an approval hierarchy and payment trail ensures accountability and compliance throughout the laytime calculation process. Users have the flexibility to define delegation of authority, and upon completion of the approval flow, the application seamlessly transitions to the payment approval process. The inclusion of features like multiple vessel calculation selection and supplier validation further enhances clarity and accuracy in operations.

Future enhancements could involve exploring integration with Account Payable processes for payment processing post-NFA, considering Phase-II implementation. This integration would enhance the application's functionality by automating and streamlining the payment approval process, reducing manual effort and improving overall efficiency. Additionally, further refinement of the user interface based on user feedback and technological advancements can enhance user experience and accessibility. Continuous updates to regulatory compliance and industry standards will ensure that the application remains relevant and effective in meeting the evolving needs of vessel operations management.

REFERENCES

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